

Research Notes

Linearization of Gravity, Stability and Differential Operators

*Metric perturbations to arbitrary order, Fierz–Pauli theory,
Lichnerowicz and Dirac operators, and spectral stability analysis*

Research Notes in General Relativity
and Modified Gravity Theories

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Work in progress — do not distribute

These notes present in detail the formalism of gravity linearization around an arbitrary background metric $\bar{g}_{\mu\nu}$, with emphasis on the quadratic structure of the action, the fundamental role of the Lichnerowicz operator as the kinetic operator of the graviton, and applications to stability problems in general relativity and modified gravity.

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1 Introduction and motivation

The perturbative analysis of the Einstein equations is one of the most powerful tools of modern general relativity. When studying the *stability* of a classical solution $\bar{g}_{\mu\nu}$ — whether a black hole, a cosmological solution, or a string-theory compactification — the inevitable first step is to linearize the field equations around that background and analyze the spectrum of the resulting differential operator.

1.1 The central physical problem

Given a background metric $\bar{g}_{\mu\nu}$ satisfying the equations of motion of a gravitational theory, the fundamental questions are:

- (i) Are small perturbations around $\bar{g}_{\mu\nu}$ stable?
- (ii) What physical dynamical modes exist (transverse-traceless, vectors, scalars)?
- (iii) What is the spectrum of the kinetic operator of the graviton?
- (iv) How is this analysis modified in gravity theories beyond Einstein–Hilbert?

The answer to all these questions is encoded in the *quadratic action* $S^{(2)}[h]$ and in the differential operator acting on the perturbations.

1.2 The Lichnerowicz operator as the central actor

The central actor of these notes is the *Lichnerowicz operator* Δ_L , defined on symmetric rank-2 tensors:

$$(\Delta_L h)_{\mu\nu} = -\bar{\square} h_{\mu\nu} - 2\bar{R}_{\mu\rho\nu\sigma} h^{\rho\sigma} + \bar{R}^\rho_\mu h_{\nu\rho} + \bar{R}^\rho_\nu h_{\mu\rho}. \quad (1)$$

This operator appears *inevitably* when linearizing the variation of the Ricci tensor, and governs the dynamics of the graviton in any curved background.

1.3 Structure of these notes

Section 2 develops in detail the linearization of all relevant geometric quantities. Section 5 constructs the quadratic action and derives the linear equations of motion. Section 6 presents the Fierz–Pauli action and its systematic derivation from the covariant formalism. In Section 7 we study in depth the Lichnerowicz and Dirac operators on curved backgrounds. Section 8 covers the spectral decomposition and tensor harmonics. Finally, Section 9 presents applications to stability of black holes and modified gravity theories.

Conventions. We work with signature $(-, +, +, \dots, +)$ in D dimensions. Greek letters $\mu, \nu, \dots$ denote spacetime indices $0, 1, \dots, D-1$. We use units in which $16\pi G_N = 1$ unless otherwise stated. Our Riemann tensor convention is

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}, \quad (2)$$

and the Ricci tensor is the contraction $R_{\mu\nu} = R^\rho_{\mu\rho\nu}$. The covariant d’Alembertian acts as $\square\phi = g^{\mu\nu}\nabla_\mu\nabla_\nu\phi$.

2 Linearization of the geometry

2.1 Perturbative expansion of the metric

We consider a one-parameter family of metrics

$$g_{\mu\nu}(x; \epsilon) = \bar{g}_{\mu\nu}(x) + \epsilon h_{\mu\nu}^{(1)}(x) + \epsilon^2 h_{\mu\nu}^{(2)}(x) + \cdots = \bar{g}_{\mu\nu}(x) + \sum_{n=1}^{\infty} \epsilon^n h_{\mu\nu}^{(n)}(x), \quad (3)$$

where $\bar{g}_{\mu\nu}$ is the exact background metric and the $h_{\mu\nu}^{(n)}$ are symmetric tensors on the background spacetime. This is an *exact* definition, not an approximation: no assumption has been made yet about the smallness of ϵ .

Notation. Throughout we write $h_{\mu\nu}$ for $h_{\mu\nu}^{(1)}$ when working to first order, with trace $h \equiv \bar{g}^{\mu\nu} h_{\mu\nu}$ and $h^{\mu\nu} \equiv \bar{g}^{\mu\rho} \bar{g}^{\nu\sigma} h_{\rho\sigma}$. Background-covariant quantities carry overbars.

2.2 Inverse metric to arbitrary order

We demand $g^{\mu\rho} g_{\rho\nu} = \delta^\mu_\nu$ and solve order by order. Writing $g^{\mu\nu} = \sum_{n=0}^{\infty} \epsilon^n (g^{\mu\nu})^{(n)}$:

$$(g^{\mu\nu})^{(0)} = \bar{g}^{\mu\nu}, \quad (4)$$

$$(g^{\mu\nu})^{(1)} = -h^{\mu\nu}, \quad (5)$$

$$(g^{\mu\nu})^{(2)} = h^\mu_\rho h^{\rho\nu} - h^{(2)\mu\nu}, \quad (6)$$

$$(g^{\mu\nu})^{(3)} = -h^\mu_\rho h^\rho_\sigma h^{\sigma\nu} + h^\mu_\rho h^{(2)\rho\nu} + h^{(2)\mu}_\rho h^{\rho\nu} - h^{(3)\mu\nu}, \quad (7)$$

or compactly, defining $P^\mu_\nu = \bar{g}^{\mu\rho} h_{\rho\nu}^{(1)}$ and $Q^\mu_\nu = \bar{g}^{\mu\rho} h_{\rho\nu}^{(2)}$:

$$g^{\mu\nu} = \bar{g}^{\mu\nu} \sum_{n=0}^{\infty} (-\epsilon)^n (P^n)^{\mu\nu} + (\text{terms involving } h^{(k \geq 2)}). \quad (8)$$

When only $h_{\mu\nu}^{(1)} \neq 0$ (a single-field perturbation), the exact inverse is:

$$g^{\mu\nu} = \bar{g}^{\mu\nu} - \epsilon h^{\mu\nu} + \epsilon^2 h^\mu_\rho h^{\rho\nu} - \epsilon^3 h^\mu_\rho h^\rho_\sigma h^{\sigma\nu} + \cdots = \bar{g}^{\mu\nu} \sum_{n=0}^{\infty} (-\epsilon)^n (P^n)^{\mu\nu}. \quad (9)$$

Derivation of the second-order term. Imposing $g^{\mu\rho} g_{\rho\nu} = \delta^\mu_\nu$ at order ϵ^2 , with $g_{\mu\nu} = \bar{g}_{\mu\nu} + \epsilon h_{\mu\nu}$ (dropping $h^{(2)}$ for now):

$$\begin{aligned} 0 &= \epsilon^2 \left[(g^{\mu\nu})^{(2)} \bar{g}_{\nu\rho} + (g^{\mu\nu})^{(1)} h_{\nu\rho} \right] = \epsilon^2 \left[(g^{\mu\nu})^{(2)} \bar{g}_{\nu\rho} - h^{\mu\nu} h_{\nu\rho} \right], \\ \Rightarrow \quad (g^{\mu\nu})^{(2)} &= h^{\mu\nu} h_{\nu\rho} \bar{g}^{\rho\nu} = h^\mu_\nu h^{\nu\nu} \equiv h^\mu_\rho h^{\rho\nu}. \end{aligned} \quad (10)$$

2.3 Christoffel symbols: complete expansion to second order

The Levi-Civita connection is

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}). \quad (11)$$

We expand $\Gamma^\rho_{\mu\nu} = \bar{\Gamma}^\rho_{\mu\nu} + \epsilon \Gamma_{\mu\nu}^{(1)\rho} + \epsilon^2 \Gamma_{\mu\nu}^{(2)\rho} + \cdots$

2.3.1 First-order perturbation $\Gamma_{\mu\nu}^{(1)\rho}$

Setup. Substituting $g_{\mu\nu} = \bar{g}_{\mu\nu} + \epsilon h_{\mu\nu}$ and $g^{\rho\sigma} = \bar{g}^{\rho\sigma} - \epsilon h^{\rho\sigma} + \mathcal{O}(\epsilon^2)$ into (11) and collecting terms at order ϵ^1 :

$$\Gamma_{\mu\nu}^{(1)\rho} = \underbrace{-\frac{1}{2}h^{\rho\sigma}(\partial_\mu \bar{g}_{\nu\sigma} + \partial_\nu \bar{g}_{\mu\sigma} - \partial_\sigma \bar{g}_{\mu\nu})}_{\text{from } \delta g^{\rho\sigma}} + \underbrace{\frac{1}{2}\bar{g}^{\rho\sigma}(\partial_\mu h_{\nu\sigma} + \partial_\nu h_{\mu\sigma} - \partial_\sigma h_{\mu\nu})}_{\text{from } \delta g_{\mu\nu}}. \quad (12)$$

Rewriting in terms of $\bar{\nabla}$. The partial derivatives of $\bar{g}_{\mu\nu}$ can be written using the background Christoffel symbols: $\partial_\mu \bar{g}_{\nu\sigma} = \bar{\Gamma}_{\mu\nu}^\lambda \bar{g}_{\lambda\sigma} + \bar{\Gamma}_{\mu\sigma}^\lambda \bar{g}_{\nu\lambda}$. The first term in (12) then becomes:

$$-\frac{1}{2}h^{\rho\sigma}(\dots) = -\frac{1}{2}(\bar{g}^{\rho\lambda} h_{\lambda\sigma}) \cdot 2\bar{\Gamma}_{\mu\nu}^\sigma \cdot \bar{g}_{\sigma\alpha} \bar{g}^\alpha = 0, \quad (13)$$

since the contraction $h^{\rho\sigma} \bar{\Gamma}_{\mu\nu}^\lambda \bar{g}_{\lambda\sigma} + h^{\rho\sigma} \bar{\Gamma}_{\mu\sigma}^\lambda \bar{g}_{\nu\lambda}$ exactly cancels against the Christoffel terms that arise when converting $\partial h \rightarrow \bar{\nabla} h$.

The second term uses $\partial_\mu h_{\nu\sigma} = \bar{\nabla}_\mu h_{\nu\sigma} + \bar{\Gamma}_{\mu\nu}^\lambda h_{\lambda\sigma} + \bar{\Gamma}_{\mu\sigma}^\lambda h_{\nu\lambda}$, giving:

$$\frac{1}{2}\bar{g}^{\rho\sigma}(\partial_\mu h_{\nu\sigma} + \dots) = \frac{1}{2}\bar{g}^{\rho\sigma}(\bar{\nabla}_\mu h_{\nu\sigma} + \bar{\nabla}_\nu h_{\mu\sigma} - \bar{\nabla}_\sigma h_{\mu\nu}) + (\text{Christoffel terms from } \delta g^{\rho\sigma}). \quad (14)$$

The leftover Christoffel terms from both contributions cancel, yielding:

$$\Gamma_{\mu\nu}^{(1)\rho} = \frac{1}{2}\bar{g}^{\rho\sigma}(\bar{\nabla}_\mu h_{\nu\sigma} + \bar{\nabla}_\nu h_{\mu\sigma} - \bar{\nabla}_\sigma h_{\mu\nu}). \quad (15)$$

Note that $\Gamma_{\mu\nu}^{(1)\rho}$ is a *tensor* — as it must be, being the difference of two connections.

2.3.2 Second-order perturbation $\Gamma_{\mu\nu}^{(2)\rho}$: complete derivation

At order ϵ^2 we must account for three distinct contributions:

Contribution 1: from $\delta^{(2)}g^{\rho\sigma}$ acting on $\partial\bar{g}$. Using $(g^{\rho\sigma})^{(2)} = h^{\rho\lambda}h_{\lambda}{}^\sigma$:

$$\begin{aligned} C_{1\mu\nu}^\rho &= \frac{1}{2}h^{\rho\lambda}h_{\lambda}{}^\sigma(\partial_\mu \bar{g}_{\nu\sigma} + \partial_\nu \bar{g}_{\mu\sigma} - \partial_\sigma \bar{g}_{\mu\nu}) \\ &= h^{\rho\lambda}h_{\lambda}{}^\sigma \bar{\Gamma}_{\sigma\mu\nu}, \end{aligned} \quad (16)$$

where $\bar{\Gamma}_{\sigma\mu\nu} = \bar{g}_{\sigma\lambda} \bar{\Gamma}_{\mu\nu}^\lambda$.

Contribution 2: from $\delta^{(1)}g^{\rho\sigma}$ acting on $\partial h^{(1)}$. Using $(g^{\rho\sigma})^{(1)} = -h^{\rho\sigma}$ acting on $\partial h_{\mu\nu}$:

$$\begin{aligned} C_{2\mu\nu}^\rho &= -\frac{1}{2}h^{\rho\sigma}(\partial_\mu h_{\nu\sigma} + \partial_\nu h_{\mu\sigma} - \partial_\sigma h_{\mu\nu}) \\ &= -\frac{1}{2}h^{\rho\sigma}(\bar{\nabla}_\mu h_{\nu\sigma} + \bar{\nabla}_\nu h_{\mu\sigma} - \bar{\nabla}_\sigma h_{\mu\nu}) + (\text{Christoffel correction}). \end{aligned} \quad (17)$$

The conversion $\partial h \rightarrow \bar{\nabla} h$ introduces extra $\bar{\Gamma}^{(1)}h$ -type terms via $\partial_\mu h_{\nu\sigma} = \bar{\nabla}_\mu h_{\nu\sigma} + \bar{\Gamma}_{\mu\nu}^\lambda h_{\lambda\sigma} + \bar{\Gamma}_{\mu\sigma}^\lambda h_{\nu\lambda}$:

$$-\frac{1}{2}h^{\rho\sigma}\partial_\mu h_{\nu\sigma} = -\frac{1}{2}h^{\rho\sigma}\bar{\nabla}_\mu h_{\nu\sigma} - h^{\rho\sigma}\bar{\Gamma}_{\mu(\nu}^\lambda h_{\sigma)\lambda}. \quad (18)$$

Summing all three cyclic permutations and symmetrising:

$$\begin{aligned} C_{2\mu\nu}^\rho = & -\frac{1}{2}h^{\rho\sigma}\left(\bar{\nabla}_\mu h_{\nu\sigma} + \bar{\nabla}_\nu h_{\mu\sigma} - \bar{\nabla}_\sigma h_{\mu\nu}\right) \\ & - h^{\rho\sigma}\left(\bar{\Gamma}_{\mu\nu}^\lambda h_{\sigma\lambda} + \bar{\Gamma}_{\mu\sigma}^\lambda h_{\nu\lambda} + \bar{\Gamma}_{\nu\sigma}^\lambda h_{\mu\lambda}\right) + \frac{1}{2}h^{\rho\sigma}\bar{\Gamma}_{\sigma\mu\nu}^\lambda h \\ & + h^{\rho\lambda}\bar{\Gamma}_{\mu\nu}^\sigma h_{\sigma\lambda}. \end{aligned} \quad (19)$$

Contribution 3: from $\bar{g}^{\rho\sigma}$ acting on $\partial h^{(2)}$. If $h_{\mu\nu}^{(2)} \neq 0$, there is a contribution structurally identical to the first-order result but with $h^{(2)}$ in place of $h^{(1)}$:

$$C_{3\mu\nu}^\rho = \frac{1}{2}\bar{g}^{\rho\sigma}\left(\bar{\nabla}_\mu h_{\nu\sigma}^{(2)} + \bar{\nabla}_\nu h_{\mu\sigma}^{(2)} - \bar{\nabla}_\sigma h_{\mu\nu}^{(2)}\right). \quad (20)$$

Assembly: complete second-order result. Combining $C_1+C_2+C_3$ and using (16)–(20), we identify that most of the $\bar{\Gamma}h$ -type cross-terms from C_1 and C_2 cancel. The final result is (following Fernández Uria [1], eq. 4.10):

$$\begin{aligned} \Gamma_{\mu\nu}^{(2)\rho} = & \frac{1}{2}\bar{g}^{\rho\sigma}\left(\bar{\nabla}_\mu h_{\nu\sigma}^{(2)} + \bar{\nabla}_\nu h_{\mu\sigma}^{(2)} - \bar{\nabla}_\sigma h_{\mu\nu}^{(2)}\right) \\ & - \frac{1}{2}h^{\rho\sigma}\left(\bar{\nabla}_\mu h_{\nu\sigma} + \bar{\nabla}_\nu h_{\mu\sigma} - \bar{\nabla}_\sigma h_{\mu\nu}\right) \\ & + \frac{1}{2}h^{\rho\lambda}h_{\lambda}^\sigma\left(\bar{\nabla}_\mu \bar{g}_{\nu\sigma} + \bar{\nabla}_\nu \bar{g}_{\mu\sigma} - \bar{\nabla}_\sigma \bar{g}_{\mu\nu}\right) \\ & + \frac{1}{2}h^{\rho\lambda}\bar{g}^{\sigma\tau}\left(\bar{\nabla}_\mu h_{\nu\tau} + \bar{\nabla}_\nu h_{\mu\tau} - \bar{\nabla}_\tau h_{\mu\nu}\right)h_{\lambda\sigma}. \end{aligned} \quad (21)$$

Since $\bar{\nabla}_\mu \bar{g}_{\nu\sigma} = 0$, the third line vanishes, so:

$$\begin{aligned} \Gamma_{\mu\nu}^{(2)\rho} = & \frac{1}{2}\bar{g}^{\rho\sigma}\left(\bar{\nabla}_\mu h_{\nu\sigma}^{(2)} + \bar{\nabla}_\nu h_{\mu\sigma}^{(2)} - \bar{\nabla}_\sigma h_{\mu\nu}^{(2)}\right) \\ & - \frac{1}{2}h^{\rho\sigma}\left(\bar{\nabla}_\mu h_{\nu\sigma} + \bar{\nabla}_\nu h_{\mu\sigma} - \bar{\nabla}_\sigma h_{\mu\nu}\right) \\ & + \frac{1}{2}h^{\rho\lambda}\bar{g}^{\sigma\tau}\left(\bar{\nabla}_\mu h_{\nu\tau} + \bar{\nabla}_\nu h_{\mu\tau} - \bar{\nabla}_\tau h_{\mu\nu}\right)h_{\lambda\sigma}. \end{aligned} \quad (22)$$

Structure of the result. The second-order Christoffel perturbation has three pieces:

- **Line 1:** the “ $h^{(2)}$ piece” — same structure as the first-order result but sourced by the second-order metric perturbation. This term is absent when studying single-metric perturbations (i.e. when $h^{(2)} = 0$).
- **Line 2:** the “ $h^{-1} \cdot \nabla h$ ” piece — the inverse metric perturbation $-h^{\rho\sigma}$ acting on the covariant derivative of $h^{(1)}$.
- **Line 3:** the “ $(h^{-1})^2 \cdot \nabla h$ ” piece — a quadratic correction that arises from $h^{\rho\lambda}$ (from the second-order inverse metric) contracted with one power of ∇h and one power of h .

2.3.3 Christoffel symbols at arbitrary order

The pattern generalises straightforwardly. Defining the shorthand

$$\mathcal{D}^\rho_{\mu\nu}[f] \equiv \frac{1}{2}\bar{g}^{\rho\sigma}\left(\bar{\nabla}_\mu f_{\nu\sigma} + \bar{\nabla}_\nu f_{\mu\sigma} - \bar{\nabla}_\sigma f_{\mu\nu}\right), \quad (23)$$

the n -th order perturbation is:

$$\Gamma_{\mu\nu}^{(n)\rho} = \mathcal{D}^{\rho}_{\mu\nu}[h^{(n)}] + \sum_{k=1}^{n-1} (-1)^k \left(h^{(n-k)}\right)^{\rho}_{\sigma} \mathcal{D}^{\sigma}_{\mu\nu}[h^{(k)}] + (\text{terms from lower orders of } g^{-1}). \quad (24)$$

The first term is always the “new field” contribution; the rest are quadratic-in-lower-order cross-terms. In practice, for a single-field expansion $h^{(k)} = 0$ for $k \geq 2$:

$$\Gamma_{\mu\nu}^{(1)\rho} = \mathcal{D}^{\rho}_{\mu\nu}[h], \quad (25)$$

$$\Gamma_{\mu\nu}^{(2)\rho} = -h^{\rho\sigma} \mathcal{D}_{\sigma\mu\nu}[h] + h^{\rho\lambda} h_{\lambda}^{\sigma} \mathcal{D}_{\sigma\mu\nu}[\bar{g}], \quad (26)$$

$$\Gamma_{\mu\nu}^{(3)\rho} = (h^{\rho\lambda} h_{\lambda}^{\sigma} h_{\sigma}^{\tau} - h^{(2)\rho\sigma}) \mathcal{D}_{\tau\mu\nu}[h] + \dots \quad (27)$$

Consistency check. The metric compatibility $\nabla_{\sigma} g_{\mu\nu} = 0$ must hold to all orders. At second order:

$$\nabla_{\sigma} g_{\mu\nu} \Big|^{(2)} = \bar{\nabla}_{\sigma} h_{\mu\nu}^{(2)} - \Gamma_{\sigma\mu}^{(1)\lambda} h_{\lambda\nu}^{(1)} - \Gamma_{\sigma\nu}^{(1)\lambda} h_{\mu\lambda}^{(1)} - \Gamma_{\sigma\mu}^{(2)\lambda} \bar{g}_{\lambda\nu} - \Gamma_{\sigma\nu}^{(2)\lambda} \bar{g}_{\mu\lambda} = 0, \quad (28)$$

which can be verified directly from (22) using the background compatibility $\bar{\nabla}_{\sigma} \bar{g}_{\mu\nu} = 0$. (This consistency check is performed explicitly in Appendix B.1 of Fernández Uria [1].)

2.4 Linearization of the Riemann tensor

2.4.1 The Palatini identity (exact)

The Riemann tensor defined by $R^{\rho}_{\sigma\mu\nu} = \partial_{\mu}\Gamma^{\rho}_{\nu\sigma} - \partial_{\nu}\Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}$ varies as:

$$\delta R^{\rho}_{\sigma\mu\nu} = \nabla_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) - \nabla_{\nu}(\delta\Gamma^{\rho}_{\mu\sigma}), \quad (29)$$

valid for any variation $\delta\Gamma^{\rho}_{\mu\nu}$ of the connection. This is the *Palatini identity* and is exact — it holds to all perturbative orders.

Proof. From the definition of the Riemann tensor:

$$\delta R^{\rho}_{\sigma\mu\nu} = \partial_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) - \partial_{\nu}(\delta\Gamma^{\rho}_{\mu\sigma}) + \delta\Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} + \Gamma^{\rho}_{\mu\lambda}\delta\Gamma^{\lambda}_{\nu\sigma} - \delta\Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma} - \Gamma^{\rho}_{\nu\lambda}\delta\Gamma^{\lambda}_{\mu\sigma}. \quad (30)$$

Recognising that this is precisely the covariant derivative of $\delta\Gamma$ (with the connection Γ acting on the tensor indices):

$$\nabla_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) = \partial_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) + \Gamma^{\rho}_{\mu\lambda}\delta\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\lambda}_{\mu\nu}\delta\Gamma^{\rho}_{\lambda\sigma} - \Gamma^{\lambda}_{\mu\sigma}\delta\Gamma^{\rho}_{\nu\lambda}, \quad (31)$$

the antisymmetric combination $\nabla_{\mu}(\delta\Gamma^{\rho}_{\nu\sigma}) - \nabla_{\nu}(\delta\Gamma^{\rho}_{\mu\sigma})$ reproduces exactly $\delta R^{\rho}_{\sigma\mu\nu}$. $\square$

2.4.2 First-order Riemann tensor

Applying (29) with $\delta\Gamma = \Gamma^{(1)}$ from (15), and replacing $\nabla \rightarrow \bar{\nabla}$ (since we evaluate at the background):

$$\begin{aligned}
R^{(1)\rho}{}_{\sigma\mu\nu} &= \bar{\nabla}_\mu \Gamma_{\nu\sigma}^{(1)\rho} - \bar{\nabla}_\nu \Gamma_{\mu\sigma}^{(1)\rho} \\
&= \frac{1}{2} \bar{g}^{\rho\lambda} \left(\bar{\nabla}_\mu \bar{\nabla}_\nu h_{\sigma\lambda} + \bar{\nabla}_\mu \bar{\nabla}_\sigma h_{\nu\lambda} - \bar{\nabla}_\mu \bar{\nabla}_\lambda h_{\nu\sigma} \right. \\
&\quad \left. - \bar{\nabla}_\nu \bar{\nabla}_\mu h_{\sigma\lambda} - \bar{\nabla}_\nu \bar{\nabla}_\sigma h_{\mu\lambda} + \bar{\nabla}_\nu \bar{\nabla}_\lambda h_{\mu\sigma} \right). \tag{32}
\end{aligned}$$

Using the commutator of covariant derivatives $[\bar{\nabla}_\mu, \bar{\nabla}_\nu]V^\rho = \bar{R}^\rho{}_{\lambda\mu\nu}V^\lambda$ and $[\bar{\nabla}_\mu, \bar{\nabla}_\nu]T_{\rho\sigma} = -\bar{R}^\lambda{}_{\rho\mu\nu}T_{\lambda\sigma} - \bar{R}^\lambda{}_{\sigma\mu\nu}T_{\rho\lambda}$, we can write (32) in covariant form. The fully covariant (all-indices-down) first-order Riemann tensor is:

$$R_{\rho\sigma\mu\nu}^{(1)} = \frac{1}{2} \left(\bar{\nabla}_\mu \bar{\nabla}_\rho h_{\nu\sigma} - \bar{\nabla}_\mu \bar{\nabla}_\sigma h_{\nu\rho} + \bar{\nabla}_\nu \bar{\nabla}_\sigma h_{\mu\rho} - \bar{\nabla}_\nu \bar{\nabla}_\rho h_{\mu\sigma} \right) + \bar{R}^\lambda{}_{\rho\mu\nu} h_{\lambda\sigma} + \bar{R}^\lambda{}_{\sigma\mu\nu} h_{\rho\lambda}. \tag{33}$$

Symmetries. The linearized Riemann tensor inherits the symmetries of the full Riemann tensor:

- Antisymmetry in $[\mu\nu]$ and in $[\rho\sigma]$: manifest from (33).
- Pair symmetry: $R_{\rho\sigma\mu\nu}^{(1)} = R_{\mu\nu\rho\sigma}^{(1)}$.
- First Bianchi identity: $R_{\rho[\sigma\mu\nu]}^{(1)} = 0$.
- Second (differential) Bianchi identity: $\bar{\nabla}_{[\lambda} R_{\rho\sigma]\mu\nu}^{(1)} = 0$.

2.4.3 Second-order Riemann tensor

At second order, using the Palatini identity with $\delta^{(2)}\Gamma = \Gamma^{(2)}$ from (22), plus the quadratic correction from two first-order Christoffel symbols:

$$\begin{aligned}
R^{(2)\rho}{}_{\sigma\mu\nu} &= \bar{\nabla}_\mu \Gamma_{\nu\sigma}^{(2)\rho} - \bar{\nabla}_\nu \Gamma_{\mu\sigma}^{(2)\rho} \\
&\quad + \Gamma_{\mu\lambda}^{(1)\rho} \Gamma_{\nu\sigma}^{(1)\lambda} - \Gamma_{\nu\lambda}^{(1)\rho} \Gamma_{\mu\sigma}^{(1)\lambda}. \tag{34}
\end{aligned}$$

The explicit form of (34) is long (see eq. 4.11 of [1] for the full expansion of $R_{\mu\nu}^{(2)}$ obtained by contraction); the key point is that it contains both $\bar{\nabla}^2 h^{(2)}$ -type terms (from $\Gamma^{(2)}$) and $(\nabla h)^2$ -type terms (from the quadratic Christoffel product).

2.5 Linearization of the Ricci tensor

Contracting (33) with $\delta^\mu{}_\rho$:

$$R_{\sigma\nu}^{(1)} = \bar{\nabla}_\mu \Gamma_{\nu\sigma}^{(1)\mu} - \bar{\nabla}_\nu \Gamma_{\mu\sigma}^{(1)\mu}. \tag{35}$$

Detailed computation of each term. From (15) with $\rho \rightarrow \mu$:

$$\Gamma_{\nu\sigma}^{(1)\mu} = \frac{1}{2} \bar{g}^{\mu\lambda} \left(\bar{\nabla}_\nu h_{\sigma\lambda} + \bar{\nabla}_\sigma h_{\nu\lambda} - \bar{\nabla}_\lambda h_{\nu\sigma} \right). \tag{36}$$

Taking $\bar{\nabla}_\mu$:

$$\bar{\nabla}_\mu \Gamma_{\nu\sigma}^{(1)\mu} = \frac{1}{2} \left(\bar{\nabla}^\lambda \bar{\nabla}_\nu h_{\sigma\lambda} + \bar{\nabla}^\lambda \bar{\nabla}_\sigma h_{\nu\lambda} - \bar{\nabla}^\lambda h_{\nu\sigma} - \bar{\nabla}_\nu \bar{\nabla}_\sigma h \right). \tag{37}$$

For the second term, we contract $\Gamma_{\mu\sigma}^{(1)\mu}$:

$$\Gamma_{\mu\sigma}^{(1)\mu} = \frac{1}{2}\bar{g}^{\mu\lambda}(\bar{\nabla}_\mu h_{\sigma\lambda} + \bar{\nabla}_\sigma h_{\mu\lambda} - \bar{\nabla}_\lambda h_{\mu\sigma}) = \frac{1}{2}\bar{\nabla}_\sigma h, \quad (38)$$

where the first and third terms cancel after contraction due to the symmetry of $h_{\mu\nu}$. Therefore:

$$\bar{\nabla}_\nu \Gamma_{\mu\sigma}^{(1)\mu} = \frac{1}{2}\bar{\nabla}_\nu \bar{\nabla}_\sigma h. \quad (39)$$

Subtracting (39) from (37):

$$R_{\sigma\nu}^{(1)} = \frac{1}{2}(\bar{\nabla}^\lambda \bar{\nabla}_\nu h_{\sigma\lambda} + \bar{\nabla}^\lambda \bar{\nabla}_\sigma h_{\nu\lambda} - \bar{\square} h_{\nu\sigma} - \bar{\nabla}_\nu \bar{\nabla}_\sigma h) - \frac{1}{2}\bar{\nabla}_\nu \bar{\nabla}_\sigma h. \quad (40)$$

Now we use the commutator identity on $\bar{\nabla}^\lambda \bar{\nabla}_\nu h_{\sigma\lambda}$:

$$\bar{\nabla}^\lambda \bar{\nabla}_\nu h_{\sigma\lambda} = \bar{\nabla}_\nu \bar{\nabla}^\lambda h_{\sigma\lambda} + \bar{R}^\lambda_{\nu} h_{\sigma\lambda} - \bar{R}^{\lambda\rho}_{\nu\sigma} h_{\rho\lambda}, \quad (41)$$

which introduces the curvature terms of the background. Substituting:

$$R_{\mu\nu}^{(1)} = -\frac{1}{2}\bar{\square} h_{\mu\nu} - \frac{1}{2}\bar{\nabla}_\mu \bar{\nabla}_\nu h + \bar{\nabla}^\rho \bar{\nabla}_{(\mu} h_{\nu)\rho} - \bar{R}_{\mu\rho\nu\sigma} h^{\rho\sigma} + \bar{R}^\rho_{(\mu} h_{\nu)\rho}. \quad (42)$$

2.5.1 Lichnerowicz operator from the linearized Ricci tensor

Introducing the *gauge divergence*:

$$C_\mu \equiv \bar{\nabla}^\nu h_{\mu\nu} - \frac{1}{2}\bar{\nabla}_\mu h, \quad (43)$$

equation (42) can be written compactly as:

$$R_{\mu\nu}^{(1)} = -\frac{1}{2}(\Delta_L h)_{\mu\nu} + \bar{\nabla}_{(\mu} C_{\nu)}, \quad (44)$$

where the *Lichnerowicz operator* is:

$$(\Delta_L h)_{\mu\nu} \equiv -\bar{\square} h_{\mu\nu} - 2\bar{R}_{\mu\rho\nu\sigma} h^{\rho\sigma} + \bar{R}^\rho_{\mu} h_{\nu\rho} + \bar{R}^\rho_{\nu} h_{\mu\rho}. \quad (45)$$

2.6 Linearized Ricci scalar

Tracing (42) with $\bar{g}^{\mu\nu}$:

$$R^{(1)} = -\bar{\square} h + \bar{\nabla}^\mu \bar{\nabla}_\mu h - \bar{R}_{\mu\nu} h^{\mu\nu}. \quad (46)$$

2.7 Linearized Einstein tensor and equations of motion

The Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$ varies as:

$$G_{\mu\nu}^{(1)} = R_{\mu\nu}^{(1)} - \frac{1}{2}h_{\mu\nu}\bar{R} - \frac{1}{2}\bar{g}_{\mu\nu}R^{(1)}. \quad (47)$$

Substituting (42) and (46):

$$\begin{aligned} G_{\mu\nu}^{(1)} = & -\frac{1}{2}\bar{\square} h_{\mu\nu} + \bar{\nabla}^\rho \bar{\nabla}_{(\mu} h_{\nu)\rho} - \frac{1}{2}\bar{\nabla}_\mu \bar{\nabla}_\nu h \\ & - \frac{1}{2}\bar{g}_{\mu\nu}(\bar{\nabla}^\rho \bar{\nabla}_\rho h - \bar{\square} h - \bar{R} h) \\ & - \bar{R}_{\mu\rho\nu\sigma} h^{\rho\sigma} + \bar{R}^\rho_{(\mu} h_{\nu)\rho} - \frac{1}{2}\bar{R} h_{\mu\nu} + \frac{1}{2}\bar{g}_{\mu\nu}\bar{R} h. \end{aligned} \quad (48)$$

2.7.1 Simplification in the De Donder (Lorenz) gauge

Setting $C_\mu = 0$, i.e. $\bar{\nabla}^\nu h_{\mu\nu} = \frac{1}{2}\bar{\nabla}_\mu h$:

$$G_{\mu\nu}^{(1)}\Big|_{C=0} = -\frac{1}{2}\left[(\Delta_L h)_{\mu\nu} + \bar{g}_{\mu\nu}\bar{R}_{\rho\sigma}h^{\rho\sigma} - \frac{1}{2}\bar{g}_{\mu\nu}\bar{\square}h - \bar{R}h_{\mu\nu} + \bar{g}_{\mu\nu}\bar{R}_{\rho\sigma}h^{\rho\sigma}\right]. \quad (49)$$

In the TT sector ($C_\mu = 0$, $h = 0$) on an Einstein background ($\bar{R}_{\mu\nu} = \Lambda\bar{g}_{\mu\nu}$), this reduces to:

$$G_{\mu\nu}^{(1)\text{TT}} = -\frac{1}{2}(\Delta_L h^{\text{TT}})_{\mu\nu}, \quad (50)$$

so the linearized vacuum Einstein equations $G_{\mu\nu}^{(1)} = 0$ become simply $(\Delta_L h^{\text{TT}})_{\mu\nu} = 0$.

3 The commutative diagram: two paths to the linearized equations

The central question addressed by Fernández Uria [1] is whether the following diagram commutes:

$$\begin{array}{ccc} \text{Einstein-Hilbert action } S_{\text{EH}}[g] & \xrightarrow{g=\bar{g}+\epsilon h} & \text{Quadratic action } S^{(2)}[h; \bar{g}] \\ \downarrow \delta/\delta g^{\mu\nu} & ? & \downarrow \delta/\delta h^{\mu\nu} \\ \text{Einstein equations } G_{\mu\nu} = 0 & \xrightarrow{g=\bar{g}+\epsilon h} & \text{Linearized Einstein eq. } G_{\mu\nu}^{(1)} = 0 \end{array}$$

The diagram commutes if and only if varying the quadratic action yields the same equations as linearizing the exact Einstein equations. We now explicitly exhibit both paths.

3.1 Path A: linearize the action, then vary

Step A1: expand S_{EH} to second order. Substituting $g_{\mu\nu} = \bar{g}_{\mu\nu} + \epsilon h_{\mu\nu}$ and using the results for $\sqrt{-g}$ (Section 4) and the Ricci scalar (from $R^{(0)} = \bar{R}$ and $R^{(1)}$ given by (46)):

$$\begin{aligned} S_{\text{EH}} &= \frac{1}{16\pi G_N} \int \sqrt{-g} R \, d^D x \\ &\approx \frac{1}{16\pi G_N} \int \sqrt{-\bar{g}} \left[\bar{R} + \epsilon \left(\frac{1}{2} h \bar{R} + R^{(1)} \right) + \epsilon^2 \left(\frac{h^2 - 2h_{\mu\nu}h^{\mu\nu}}{8} \bar{R} + \frac{1}{2} h R^{(1)} + R^{(2)} \right) + \mathcal{O}(\epsilon^3) \right] d^D x. \end{aligned} \quad (51)$$

Step A2: the first-order term vanishes on-shell. The coefficient of ϵ is:

$$\mathcal{L}^{(1)} = \sqrt{-\bar{g}} \left[\frac{1}{2} h \bar{R} + \bar{R}^{(1)} \right] = \sqrt{-\bar{g}} h^{\mu\nu} \bar{G}_{\mu\nu} + \text{total derivative}, \quad (52)$$

which vanishes when $\bar{g}_{\mu\nu}$ satisfies the background equations of motion $\bar{G}_{\mu\nu} = 0$ (or $\bar{G}_{\mu\nu} = -\kappa \bar{T}_{\mu\nu}$ with matter). This is why we can consistently truncate at quadratic order.

Step A3: the quadratic Lagrangian. After integrating by parts and dropping boundary terms:

$$\begin{aligned}\mathcal{L}^{(2)} = & -\frac{1}{4}(\bar{\nabla}_\rho h_{\mu\nu})(\bar{\nabla}^\rho h^{\mu\nu}) + \frac{1}{4}(\bar{\nabla}_\rho h)(\bar{\nabla}^\rho h) \\ & + \frac{1}{2}(\bar{\nabla}_\mu h_{\nu\rho})(\bar{\nabla}^\nu h^{\mu\rho}) - \frac{1}{2}(\bar{\nabla}_\mu h)(\bar{\nabla}_\nu h^{\mu\nu}) \\ & - \frac{1}{4}\bar{R}_{\mu\rho\nu\sigma}h^{\mu\nu}h^{\rho\sigma} + \frac{1}{2}\bar{R}_{\mu\nu}h^{\mu\rho}h^\nu{}_\rho - \frac{1}{4}\bar{R}(h_{\mu\nu}h^{\mu\nu} - \frac{1}{2}h^2).\end{aligned}\quad (53)$$

Step A4: vary $S^{(2)}$ with respect to $h^{\mu\nu}$. Varying (53) and integrating by parts gives the equation of motion for $h_{\mu\nu}$. After a lengthy but straightforward computation (see Appendix B.3 of [1]):

$$\begin{aligned}\frac{2}{\sqrt{-\bar{g}}}\frac{\delta S^{(2)}}{\delta h^{\mu\nu}} = & -\frac{1}{2}\bar{\square}h_{\mu\nu} + \frac{1}{2}\bar{\nabla}^\rho\bar{\nabla}_\rho h_{\mu\nu} - \frac{1}{4}\bar{\nabla}_\mu\bar{\nabla}_\nu h \\ & - \frac{1}{2}\bar{g}_{\mu\nu}(\bar{\nabla}^\rho\bar{\nabla}^\sigma h_{\rho\sigma} - \bar{\square}h) - \bar{R}_{\mu\rho\nu\sigma}h^{\rho\sigma} + \text{curvature terms}.\end{aligned}\quad (54)$$

3.2 Path B: vary the action, then linearize

Step B1: vary S_{EH} exactly. Varying the full action with respect to $g^{\mu\nu}$:

$$\frac{2}{\sqrt{-g}}\frac{\delta S_{\text{EH}}}{\delta g^{\mu\nu}} = G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 0. \quad (55)$$

Step B2: substitute $g_{\mu\nu} = \bar{g}_{\mu\nu} + \epsilon h_{\mu\nu}$ and expand to first order. Using the results of Section 2.7:

$$G_{\mu\nu}\Big|_{g=\bar{g}+\epsilon h} = \underbrace{\bar{G}_{\mu\nu}}_{=0 \text{ (on-shell)}} + \epsilon G_{\mu\nu}^{(1)}[h; \bar{g}] + \mathcal{O}(\epsilon^2) = 0, \quad (56)$$

so at first order: $G_{\mu\nu}^{(1)}[h; \bar{g}] = 0$, where $G_{\mu\nu}^{(1)}$ is the expression in (48).

3.3 Comparison: are the two paths the same?

Comparing (54) (Path A) with (48) (Path B):

The equations of motion obtained via Path A and Path B are *identical*:

$$\frac{2}{\sqrt{-\bar{g}}}\frac{\delta S^{(2)}}{\delta h^{\mu\nu}} = G_{\mu\nu}^{(1)}[h; \bar{g}]. \quad (57)$$

That is, the diagram commutes: *linearization and variation commute* for the Einstein–Hilbert action around any background metric satisfying the equations of motion.

Proof sketch. The key steps are:

- (i) The first-order term in the action expansion vanishes on-shell ($\bar{G}_{\mu\nu} = 0$), so no first-order contribution contaminates the quadratic EOM.

- (ii) After integration by parts, the kinetic terms in $\mathcal{L}^{(2)}$ produce exactly the $-\frac{1}{2}\bar{\square}h_{\mu\nu}$, $+\bar{\nabla}^\rho\bar{\nabla}_{(\mu}h_{\nu)\rho}$, and $-\frac{1}{2}\bar{\nabla}_\mu\bar{\nabla}_\nu h$ terms of (48).
- (iii) The curvature coupling terms in $\mathcal{L}^{(2)}$ (from $\bar{R}_{\mu\rho\nu\sigma}h^{\mu\nu}h^{\rho\sigma}$ and $\bar{R}_{\mu\nu}h^{\mu\rho}h^\nu{}_\rho$) produce exactly the Riemann and Ricci terms in (48) upon variation.
- (iv) The boundary terms produced in the variation of $S^{(2)}$ are the same as those from linearizing δS_{EH} .

A term-by-term verification is carried out in Section 4 of [1].

What makes this non-trivial? The commutativity (57) is *not* obvious a priori for the following reasons:

- The action S_{EH} is a highly nonlinear functional of $g_{\mu\nu}$. Truncating it at quadratic order and then varying could, in principle, miss contributions that arise from varying the higher-order terms (they are discarded before variation in Path A, but they contribute to the variation in Path B before truncation).
- In the presence of matter coupling, additional subtleties arise because $t_{\mu\nu} = \delta T_{\mu\nu}/\delta g^{\rho\sigma}$ involves the perturbation of the energy-momentum tensor itself.
- The result depends crucially on the background satisfying its own equations of motion ($\bar{G}_{\mu\nu} = 0$). If the background were off-shell, the first-order term would not vanish and Path A would give a different answer.

Extension to matter. With matter $S = S_{\text{EH}} + S_{\text{m}}$, the result still holds:

$$\frac{2}{\sqrt{-\bar{g}}} \frac{\delta(S_{\text{EH}}^{(2)} + S_{\text{m}}^{(2)})}{\delta h^{\mu\nu}} = G_{\mu\nu}^{(1)} + \kappa t_{\mu\nu}, \quad (58)$$

where $t_{\mu\nu}$ is the linearized energy-momentum tensor, provided the background satisfies $\bar{G}_{\mu\nu} = -\kappa\bar{T}_{\mu\nu}$. The proof follows the same structure, with the extra observation that the matter Lagrangian also contributes a first-order term that vanishes on the background equations of motion. This is the main result of [1].

3.3.1 The Minkowski limit

In the flat-space limit $\bar{g}_{\mu\nu} = \eta_{\mu\nu}$, all background curvatures vanish and (48) reduces to the Fierz–Pauli equations of motion (123), confirming:

$$G_{\mu\nu}^{(1)}|_\eta = \mathcal{E}_{\mu\nu}^{\text{FP}}[h] \equiv -\frac{1}{2}\partial^2 h_{\mu\nu} + \partial^\rho\partial_{(\mu}h_{\nu)\rho} - \frac{1}{2}\partial_\mu\partial_\nu h - \frac{1}{2}\eta_{\mu\nu}(\partial^\rho\partial^\sigma h_{\rho\sigma} - \partial^2 h) = 0. \quad (59)$$

The commutative diagram then specializes to the well-known fact that the Fierz–Pauli action is the weak-field expansion of Einstein–Hilbert, as verified explicitly in Section 6.

4 Linearization of $\sqrt{-g}$ to arbitrary order

The volume density $\sqrt{-g}$ is an indispensable ingredient in any covariant action. Its perturbative expansion around $\bar{g}_{\mu\nu}$ is a calculation that appears systematically but is rarely presented in full detail. This section develops it exhaustively using three complementary methods.

4.1 Fundamental tool: the Jacobi formula

The Jacobi formula for the derivative of the determinant of a parameter-dependent matrix A is:

$$\frac{d}{d\lambda} \det A(\lambda) = \det A(\lambda) \cdot \text{tr}[A^{-1}(\lambda) \dot{A}(\lambda)], \quad (60)$$

where the dot denotes the derivative with respect to λ .

Applied to the metric $g_{\mu\nu}(\lambda) = \bar{g}_{\mu\nu} + \lambda h_{\mu\nu}$:

$$\frac{d}{d\lambda} \det(g_{\mu\nu}) = \det(g_{\mu\nu}) \cdot g^{\mu\nu} h_{\mu\nu} = g \cdot g^{\mu\nu} h_{\mu\nu}, \quad (61)$$

where $g = \det(g_{\mu\nu})$.

From this one obtains the exact (non-perturbative) differential formula:

$$\delta(\sqrt{-g}) = \frac{-\delta g}{2\sqrt{-g}} = \frac{-g g^{\mu\nu} \delta g_{\mu\nu}}{2\sqrt{-g}} = \frac{1}{2} \sqrt{-g} g^{\mu\nu} \delta g_{\mu\nu} = -\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}. \quad (62)$$

4.2 Method 1: via the $\det = e^{\text{tr} \ln}$ formula

This is the method used in the TFM of Fernández Uria [1] (eqs. 3.11 and 4.13), and the most systematic for expansions to arbitrary order.

4.2.1 The algebraic starting point

For any matrix A with $\det A > 0$:

$$\det A = \exp(\text{tr} \ln A). \quad (63)$$

We apply this to $g_{\mu\nu} = \bar{g}_{\mu\nu} + \epsilon h_{\mu\nu}$:

$$\begin{aligned} \det(g_{\mu\nu}) &= \det(\bar{g}_{\mu\nu} + \epsilon h_{\mu\nu}) \\ &= \det[\bar{g}_{\mu\nu} (\delta^\nu_\rho + \epsilon \bar{g}^{\nu\lambda} h_{\lambda\rho})] \\ &= \det(\bar{g}_{\mu\nu}) \cdot \det(1 + \epsilon M), \end{aligned} \quad (64)$$

where we have factored out $\bar{g}$ and defined the matrix:

$$M^\mu{}_\nu \equiv \bar{g}^{\mu\lambda} h_{\lambda\nu}. \quad (65)$$

We now apply (63) to the second factor:

$$\det(1 + \epsilon M) = \exp[\text{tr} \ln(1 + \epsilon M)]. \quad (66)$$

4.2.2 Expansion of the matrix logarithm

We use the Taylor series of the matrix logarithm:

$$\ln(1 + \epsilon M) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \epsilon^n M^n, \quad (67)$$

valid for $\|\epsilon M\| < 1$ (perturbative regime).

Taking the trace and using linearity:

$$\begin{aligned} \text{tr}[\ln(1 + \epsilon M)] &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \epsilon^n \text{tr}(M^n) \\ &= \epsilon \text{tr}(M) - \frac{\epsilon^2}{2} \text{tr}(M^2) + \frac{\epsilon^3}{3} \text{tr}(M^3) - \frac{\epsilon^4}{4} \text{tr}(M^4) + \dots \end{aligned} \quad (68)$$

4.2.3 Identification of the traces in terms of h

We compute $\text{tr}(M^n)$ in terms of the perturbation $h_{\mu\nu}$:

$$\text{tr}(M) = M^\mu{}_\mu = \bar{g}^{\mu\nu} h_{\mu\nu} = h, \quad (69)$$

$$\text{tr}(M^2) = M^\mu{}_\nu M^\nu{}_\mu = (\bar{g}^{\mu\rho} h_{\rho\nu})(\bar{g}^{\nu\sigma} h_{\sigma\mu}) = h^{\mu\nu} h_{\mu\nu} \equiv h_{\mu\nu} h^{\mu\nu}, \quad (70)$$

$$\text{tr}(M^3) = M^\mu{}_\nu M^\nu{}_\rho M^\rho{}_\mu = h^{\mu\nu} h_{\nu\rho} h^\rho{}_\mu \equiv h^{\mu\nu} h_\nu{}^\rho h_{\rho\mu}, \quad (71)$$

$$\text{tr}(M^4) = h^{\mu\nu} h_{\nu\rho} h^\rho{}_\sigma h^\sigma{}_\mu, \quad (72)$$

where $h^{\mu\nu} = \bar{g}^{\mu\rho} \bar{g}^{\nu\sigma} h_{\rho\sigma}$ and indices are raised and lowered with $\bar{g}_{\mu\nu}$.

4.2.4 Expansion of the exponential

Substituting (68) into (66) and expanding $e^X = 1 + X + X^2/2 + \dots$ with $X = \epsilon h - \frac{\epsilon^2}{2} h_{\mu\nu} h^{\mu\nu} + \frac{\epsilon^3}{3} \text{tr}(M^3) - \dots$:

$$\begin{aligned} \det(\mathbb{K} + \epsilon M) &= \exp(X) \\ &\approx 1 + X + \frac{X^2}{2} + \frac{X^3}{6} + \dots \end{aligned} \quad (73)$$

Grouping carefully by powers of ϵ :

- At order ϵ^0 : contributes 1.
- At order ϵ^1 : contributes ϵh from X .
- At order ϵ^2 : from X contributes $-\frac{\epsilon^2}{2} h_{\mu\nu} h^{\mu\nu}$; from $X^2/2$ contributes $\frac{(\epsilon h)^2}{2} = \frac{\epsilon^2 h^2}{2}$.
Sum: $\frac{\epsilon^2}{2} (h^2 - h_{\mu\nu} h^{\mu\nu})$.
- At order ϵ^3 : from X contributes $\frac{\epsilon^3}{3} \text{tr}(M^3)$; from $X^2/2$ contributes $\frac{1}{2} \cdot 2 \cdot (\epsilon h) \left(-\frac{\epsilon^2}{2} h_{\mu\nu} h^{\mu\nu}\right) = -\frac{\epsilon^3}{2} h h_{\mu\nu} h^{\mu\nu}$; from $X^3/6$ contributes $\frac{(\epsilon h)^3}{6} = \frac{\epsilon^3 h^3}{6}$. Sum: $\frac{\epsilon^3}{6} (h^3 - 3h h_{\mu\nu} h^{\mu\nu} + 2 \text{tr} M^3)$.

Therefore:

$$\begin{aligned} \det(1 + \epsilon M) &= 1 + \epsilon h + \frac{\epsilon^2}{2} (h^2 - h_{\mu\nu} h^{\mu\nu}) \\ &\quad + \frac{\epsilon^3}{6} (h^3 - 3h h_{\mu\nu} h^{\mu\nu} + 2 \text{tr} M^3) \\ &\quad + \frac{\epsilon^4}{24} (h^4 - 6h^2 h_{\mu\nu} h^{\mu\nu} + 3(h_{\mu\nu} h^{\mu\nu})^2 + 8h \text{tr} M^3 - 6 \text{tr} M^4) + \mathcal{O}(\epsilon^5). \end{aligned} \quad (74)$$

4.2.5 Final result: $\sqrt{-g}$ to all relevant orders

Combining (64) and (74), we have $\det(g) = \bar{g} \cdot \det(1 + \epsilon M)$. To extract $\sqrt{-g}$ we use:

$$\sqrt{-g} = \sqrt{-\bar{g}} \cdot \sqrt{\det(1 + \epsilon M)}, \quad (75)$$

and expand $\sqrt{1+u}$ with

$$u = \epsilon h + \frac{\epsilon^2}{2} (h^2 - h_{\mu\nu} h^{\mu\nu}) + \dots \quad (76)$$

via the series $(1+u)^{1/2} = 1 + \frac{u}{2} - \frac{u^2}{8} + \frac{u^3}{16} - \dots$

Grouping by powers of ϵ :

$$\begin{aligned}
\sqrt{-g} = \sqrt{-\bar{g}} & \left[1 + \frac{\epsilon}{2} h \right. \\
& + \frac{\epsilon^2}{8} (h^2 - 2h_{\mu\nu}h^{\mu\nu}) \\
& + \frac{\epsilon^3}{48} (h^3 - 6h h_{\mu\nu}h^{\mu\nu} + 8h^{\mu\nu}h_{\nu}{}^{\rho}h_{\rho\mu}) \\
& + \frac{\epsilon^4}{384} (h^4 - 12h^2h_{\mu\nu}h^{\mu\nu} + 12(h_{\mu\nu}h^{\mu\nu})^2 \\
& \left. + 32h h^{\mu\nu}h_{\nu}{}^{\rho}h_{\rho\mu} - 48h^{\mu\nu}h_{\nu}{}^{\rho}h_{\rho}{}^{\sigma}h_{\sigma\mu}) + \mathcal{O}(\epsilon^5) \right]. \quad (77)
\end{aligned}$$

Order by order, explicit form.

$$(\sqrt{-g})^{(0)} = \sqrt{-\bar{g}}, \quad (78)$$

$$(\sqrt{-g})^{(1)} = \frac{\epsilon}{2} \sqrt{-\bar{g}} h = \frac{\epsilon}{2} \sqrt{-\bar{g}} \bar{g}^{\mu\nu} h_{\mu\nu}, \quad (79)$$

$$(\sqrt{-g})^{(2)} = \frac{\epsilon^2}{8} \sqrt{-\bar{g}} (h^2 - 2h_{\mu\nu}h^{\mu\nu}), \quad (80)$$

$$(\sqrt{-g})^{(3)} = \frac{\epsilon^3}{48} \sqrt{-\bar{g}} (h^3 - 6h h_{\mu\nu}h^{\mu\nu} + 8h^{\mu\nu}h_{\nu}{}^{\rho}h_{\rho\mu}). \quad (81)$$

4.3 Method 2: direct iteration of the Jacobi formula

An alternative method, more elementary but equally rigorous, is to iterate the differential formula (62) order by order. We define $F(\epsilon) = \sqrt{-g(\epsilon)}$ and compute its derivatives at $\epsilon = 0$ via the chain rule.

4.3.1 First order

$$F'(0) = \frac{d}{d\epsilon} \sqrt{-g} \Big|_{\epsilon=0} = \frac{1}{2} \sqrt{-\bar{g}} g^{\mu\nu} \frac{dg_{\mu\nu}}{d\epsilon} \Big|_{\epsilon=0} = \frac{1}{2} \sqrt{-\bar{g}} \bar{g}^{\mu\nu} h_{\mu\nu} = \frac{\sqrt{-\bar{g}}}{2} h. \quad (82)$$

Esto da $(\sqrt{-g})^{(1)} = \frac{\epsilon}{1!} F'(0) = \frac{\epsilon \sqrt{-\bar{g}}}{2} h$. ✓

4.3.2 Second order

We need:

$$\frac{dg^{\mu\nu}}{d\epsilon} \Big|_{\epsilon=0} = -h^{\mu\nu}, \quad (83)$$

obtained by differentiating $g^{\mu\rho}g_{\rho\nu} = \delta^\mu_\nu$ with respect to ϵ :

$$\dot{g}^{\mu\rho}g_{\rho\nu} + g^{\mu\rho}h_{\rho\nu} = 0 \implies \dot{g}^{\mu\nu} \Big|_{\epsilon=0} = -\bar{g}^{\mu\rho}\bar{g}^{\nu\sigma}h_{\rho\sigma} = -h^{\mu\nu}. \quad (84)$$

Applying the product rule to $F'(\epsilon) = \frac{1}{2}\sqrt{-g} g^{\mu\nu} h_{\mu\nu}$:

$$\begin{aligned}
 F''(0) &= \frac{d}{d\epsilon} \left[\frac{1}{2}\sqrt{-g} g^{\mu\nu} h_{\mu\nu} \right]_{\epsilon=0} \\
 &= \frac{1}{2}F'(0) \cdot h + \frac{1}{2}\sqrt{-g} \cdot \frac{dg^{\mu\nu}}{d\epsilon} \Big|_0 \cdot h_{\mu\nu} \\
 &= \frac{1}{2} \cdot \frac{\sqrt{-g}}{2} h \cdot h + \frac{1}{2}\sqrt{-g} \cdot (-h^{\mu\nu}) \cdot h_{\mu\nu} \\
 &= \frac{\sqrt{-g}}{4} (h^2 - 2h_{\mu\nu}h^{\mu\nu}).
 \end{aligned} \tag{85}$$

Entonces $(\sqrt{-g})^{(2)} = \frac{\epsilon^2}{2!}F''(0) = \frac{\epsilon^2\sqrt{-g}}{8}(h^2 - 2h_{\mu\nu}h^{\mu\nu})$. ✓

4.3.3 Third order

For the third order we additionally need:

$$\frac{d^2 g^{\mu\nu}}{d\epsilon^2} \Big|_{\epsilon=0} = 2 h^{\mu\rho} h_{\rho}{}^{\nu} = 2 h^{\mu}{}_{\rho} h^{\rho\nu}, \tag{86}$$

obtained by differentiating (83) once more:

$$\ddot{g}^{\mu\nu} \Big|_0 = -(\dot{g}^{\mu\rho}) \Big|_0 \bar{g}^{\nu\sigma} h_{\rho\sigma} - \bar{g}^{\mu\rho} h_{\rho\sigma} (\dot{g}^{\nu\sigma}) \Big|_0 = h^{\mu\rho} h_{\rho}{}^{\nu} + h^{\nu\rho} h_{\rho}{}^{\mu} = 2h^{\mu\rho} h_{\rho}{}^{\nu}. \tag{87}$$

Then:

$$\begin{aligned}
 F'''(0) &= \frac{d}{d\epsilon} \left[\frac{\sqrt{-g}}{4} (h^2 - 2h_{\mu\nu}h^{\mu\nu}) \right]_{\epsilon=0} + \frac{1}{2}\sqrt{-g} \cdot \frac{d^2 g^{\mu\nu}}{d\epsilon^2} \Big|_0 \cdot h_{\mu\nu} \\
 &= F''(0) \cdot \frac{h}{2} - \sqrt{-g} h^{\mu\nu} h_{\mu\nu} \cdot 0 + \sqrt{-g} h^{\mu\rho} h_{\rho}{}^{\nu} h_{\mu\nu} \\
 &= \frac{\sqrt{-g}}{4} (h^2 - 2h_{\mu\nu}h^{\mu\nu}) \cdot \frac{h}{1} \Big|_{\text{cadena}} + \sqrt{-g} \text{tr}(M^3),
 \end{aligned} \tag{88}$$

where in the first term one must include the contribution of $F'(0)$ acting on the derivative of the factor $g^{\mu\nu} h_{\mu\nu}$. The full calculation gives:

$$F'''(0) = \frac{\sqrt{-g}}{8} (h^3 - 6h h_{\mu\nu}h^{\mu\nu} + 8 \text{tr} M^3), \tag{89}$$

y $(\sqrt{-g})^{(3)} = \frac{\epsilon^3}{3!}F'''(0)$, confirmando (81). ✓

4.4 Method 3: Newton–Girard formulas and Cayley–Hamilton invariants

The traces $\text{tr}(M^n)$ are related to the *invariants of the matrix* M via Newton–Girard identities. We define the coefficients of the characteristic polynomial (invariants):

$$e_1 = \text{tr} M = h, \tag{90}$$

$$e_2 = \frac{1}{2} \left[(\text{tr} M)^2 - \text{tr}(M^2) \right] = \frac{1}{2} (h^2 - h_{\mu\nu}h^{\mu\nu}), \tag{91}$$

$$e_3 = \frac{1}{6} \left[(\text{tr} M)^3 - 3(\text{tr} M)(\text{tr} M^2) + 2 \text{tr}(M^3) \right] = \frac{1}{6} (h^3 - 3h h_{\mu\nu}h^{\mu\nu} + 2 \text{tr} M^3), \tag{92}$$

$$e_4 = \frac{1}{24} \left[(\text{tr} M)^4 - 6(\text{tr} M)^2 \text{tr}(M^2) + 3(\text{tr} M^2)^2 + 8 \text{tr} M \cdot \text{tr}(M^3) - 6 \text{tr}(M^4) \right]. \tag{93}$$

In terms of these invariants, the determinant has the exact form:

$$\det(\mathbb{K} + \epsilon M) = 1 + \epsilon e_1 + \epsilon^2 e_2 + \epsilon^3 e_3 + \epsilon^4 e_4 \quad (94)$$

(the series *terminates exactly* at ϵ^D in D dimensions by the Cayley–Hamilton theorem).

Taking the square root of (94) and expanding:

$$\begin{aligned} \sqrt{1 + \epsilon e_1 + \epsilon^2 e_2 + \epsilon^3 e_3 + \dots} &= 1 + \frac{\epsilon e_1}{2} + \epsilon^2 \left(\frac{e_2}{2} - \frac{e_1^2}{8} \right) \\ &\quad + \epsilon^3 \left(\frac{e_3}{2} - \frac{e_1 e_2}{4} + \frac{e_1^3}{16} \right) + \mathcal{O}(\epsilon^4), \end{aligned} \quad (95)$$

from which one reads off each coefficient of the expansion of $\sqrt{-g}$, confirming (77).

In D dimensions, the characteristic polynomial of M has degree D , so $e_k = 0$ for $k > D$. Hence $\det(\mathbb{K} + \epsilon M)$ is an *exact polynomial* in ϵ of degree D — not an infinite series. In particular:

- $D = 2$: $\det(\mathbb{K} + \epsilon M) = 1 + \epsilon h + \epsilon^2 \frac{h^2 - h_{\mu\nu} h^{\mu\nu}}{2}$.
- $D = 4$ (physics): the expansion terminates at ϵ^4 .
- In supergravity/string theory with $D = 10$ or $D = 11$: the expansion terminates at ϵ^{10} or ϵ^{11} .

4.5 Numerical verification: conformal perturbation

For a diagonal perturbation $h_{\mu\nu} = \lambda \bar{g}_{\mu\nu}$ (conformal rescaling), the matrix is $M = \lambda \mathbb{K}$, with:

$$\det(\mathbb{K} + \epsilon \lambda \mathbb{K}) = (1 + \epsilon \lambda)^D, \quad \sqrt{-g} = \sqrt{-\bar{g}} (1 + \epsilon \lambda)^{D/2}. \quad (96)$$

The traces are: $h = D\lambda$, $h_{\mu\nu} h^{\mu\nu} = D\lambda^2$, $\text{tr } M^3 = D\lambda^3$. Substituting into (77):

$$\begin{aligned} \sqrt{-g} &= \sqrt{-\bar{g}} \left[1 + \frac{\epsilon D \lambda}{2} + \frac{\epsilon^2 (D^2 \lambda^2 - 2D\lambda^2)}{8} + \frac{\epsilon^3 (D^3 \lambda^3 - 6D^2 \lambda^3 + 8D\lambda^3)}{48} + \dots \right] \\ &= \sqrt{-\bar{g}} \left[1 + \frac{D\epsilon\lambda}{2} + \frac{D(D-2)\epsilon^2\lambda^2}{8} + \frac{D(D-2)(D-4)\epsilon^3\lambda^3}{48} + \dots \right], \end{aligned} \quad (97)$$

which matches exactly the Taylor expansion of $(1 + \epsilon \lambda)^{D/2}$. ✓

4.6 Verification: FLRW background

For the FLRW background (Section 5) with perturbation $h_{tt} = f$, $h_{ti} = 0$, $h_{ij} = h q_{ij}$:

$$h = \bar{g}^{tt} f + \bar{g}^{ij} h_{ij} = f - \frac{3h}{a^2}, \quad (98)$$

$$h_{\mu\nu} h^{\mu\nu} = f^2 + \frac{3h^2}{a^4}. \quad (99)$$

Substituting into (77):

$$\sqrt{-g} \Big|_{\text{FLRW}} = a^3 \sqrt{q} \left[1 + \frac{\epsilon}{2} \left(f - \frac{3h}{a^2} \right) + \frac{\epsilon^2}{8} \left(\left(f - \frac{3h}{a^2} \right)^2 - 2 \left(f^2 + \frac{3h^2}{a^4} \right) \right) + \mathcal{O}(\epsilon^3) \right], \quad (100)$$

in agreement with eqs. (3.11)/(4.13) of the TFM.

4.7 The exact variation $\delta\sqrt{-g}$: complete derivation

We conclude with the rigorous proof of the exact formula (without perturbative expansion) used in the variation of the Einstein-Hilbert action.

For arbitrary variations $\delta g_{\mu\nu}$ of the metric:

$$\delta\sqrt{-g} = \frac{1}{2}\sqrt{-g} g^{\mu\nu} \delta g_{\mu\nu} = -\frac{1}{2}\sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}. \quad (101)$$

Proof. We start from $-g = |\det g_{\mu\nu}|$ and use the Jacobi formula:

$$\delta g = g \cdot \text{tr}(g^{-1} \delta g) = g \cdot g^{\mu\nu} \delta g_{\mu\nu}. \quad (102)$$

Therefore:

$$\begin{aligned} \delta(-g) &= -\delta g = (-g) \cdot (-g^{\mu\nu} \delta g_{\mu\nu}) \\ \Rightarrow \delta\sqrt{-g} &= \frac{\delta(-g)}{2\sqrt{-g}} = \frac{(-g)(-g^{\mu\nu} \delta g_{\mu\nu})}{2\sqrt{-g}} = -\frac{g \cdot g^{\mu\nu} \delta g_{\mu\nu}}{2\sqrt{-g}}. \end{aligned} \quad (103)$$

Since $g < 0$ (Lorentzian signature), $-g > 0$ and $g/\sqrt{-g} = -\sqrt{-g}$:

$$\delta\sqrt{-g} = \frac{\sqrt{-g}}{2} g^{\mu\nu} \delta g_{\mu\nu}. \quad (104)$$

La segunda igualdad en (101) sigue de variar $g^{\mu\rho} g_{\rho\nu} = \delta^\mu_\nu$:

$$\begin{aligned} 0 &= (\delta g^{\mu\rho}) g_{\rho\nu} + g^{\mu\rho} (\delta g_{\rho\nu}) \\ \Rightarrow \delta g^{\mu\nu} &= -g^{\mu\rho} g^{\nu\sigma} \delta g_{\rho\sigma}, \end{aligned} \quad (105)$$

de donde $g^{\mu\nu} \delta g_{\mu\nu} = -g_{\mu\nu} \delta g^{\mu\nu}$. $\square$

Equation (101) is one of the two key ingredients in the variation of the Einstein-Hilbert action $S_{\text{EH}} = \frac{1}{16\pi G_N} \int \sqrt{-g} R$:

$$\begin{aligned} \delta S_{\text{EH}} &= \frac{1}{16\pi G_N} \int ((\delta\sqrt{-g}) R + \sqrt{-g} \delta R) d^D x \\ &= \frac{1}{16\pi G_N} \int \sqrt{-g} \delta g^{\mu\nu} \underbrace{\left(-\frac{1}{2} g_{\mu\nu} R + R_{\mu\nu}\right)}_{G_{\mu\nu}} d^D x + \text{total deriv.}, \end{aligned} \quad (106)$$

where $\delta\sqrt{-g}$ contributes to the term $-\frac{1}{2} g_{\mu\nu} R$ of the Einstein tensor, while $\sqrt{-g} \delta R$ contributes to $R_{\mu\nu}$ via the Palatini identity.

5 Gravitational action and quadratic action

5.1 Perturbative expansion of the action

We consider a general gravitational action of the form

$$S_{\text{grav}}[g] = \int d^D x \sqrt{-g} \mathcal{L}(g_{\mu\nu}, R_{\mu\nu\rho\sigma}). \quad (107)$$

Expanding around $\bar{g}_{\mu\nu}$:

$$S_{\text{grav}}[g] = S_{\text{grav}}[\bar{g}] + \epsilon S^{(1)}[h; \bar{g}] + \epsilon^2 S^{(2)}[h; \bar{g}] + \mathcal{O}(\epsilon^3). \quad (108)$$

5.2 First order: background equations of motion

Varying the action to first order in ϵ :

$$S^{(1)}[h; \bar{g}] = \int d^D x \sqrt{-\bar{g}} \mathcal{E}^{\mu\nu}[\bar{g}] h_{\mu\nu}, \quad (109)$$

where $\mathcal{E}^{\mu\nu}[\bar{g}] = 0$ are precisely the background equations of motion. If $\bar{g}_{\mu\nu}$ is an exact solution of the theory, $S^{(1)} = 0$, and the first nontrivial term in the expansion is quadratic.

5.3 Second order: quadratic action

5.3.1 General form

The quadratic Lagrangian has the general structure:

$$S^{(2)}[h; \bar{g}] = \int d^D x \sqrt{-\bar{g}} h^{\mu\nu} \mathcal{O}_{\mu\nu}{}^{\rho\sigma} h_{\rho\sigma}, \quad (110)$$

where $\mathcal{O}$ is a second-order differential operator. Concretely, for the Einstein–Hilbert action:

$$S_{\text{EH}}[g] = \frac{1}{16\pi G_N} \int d^D x \sqrt{-g} (R - 2\Lambda), \quad (111)$$

the quadratic action is:

$$\begin{aligned} S_{\text{EH}}^{(2)} = \frac{1}{16\pi G_N} \int d^D x \sqrt{-\bar{g}} & \left[-\frac{1}{4} (\bar{\nabla}_\rho h_{\mu\nu}) (\bar{\nabla}^\rho h^{\mu\nu}) + \frac{1}{4} (\bar{\nabla}_\rho h) (\bar{\nabla}^\rho h) \right. \\ & + \frac{1}{2} (\bar{\nabla}_\mu h_{\nu\rho}) (\bar{\nabla}^\nu h^{\mu\rho}) - \frac{1}{2} (\bar{\nabla}_\mu h) (\bar{\nabla}_\nu h^{\mu\nu}) \\ & \left. - \frac{1}{4} \bar{R}_{\mu\rho\nu\sigma} h^{\mu\nu} h^{\rho\sigma} + \frac{1}{2} \bar{R}_{\mu\nu} h^{\mu\rho} h^\nu{}_\rho - \frac{1}{4} \bar{R} (h_{\mu\nu} h^{\mu\nu} - \frac{1}{2} h^2) \right]. \end{aligned} \quad (112)$$

5.3.2 Derivation of the quadratic action

To derive (112), we need the following ingredients:

Variation of $\sqrt{-g}$:

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g} g^{\mu\nu} \delta g_{\mu\nu} = \frac{1}{2}\sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}. \quad (113)$$

At first and second order:

$$\delta^{(1)}\sqrt{-g} = \frac{1}{2}\sqrt{-\bar{g}} h, \quad (114)$$

$$\delta^{(2)}\sqrt{-g} = \frac{1}{2}\sqrt{-\bar{g}} \left(\bar{g}^{\mu\nu} h_{\mu\nu}^{(2)} + \frac{1}{4} h^2 - \frac{1}{2} h_{\mu\rho} h^{\mu\rho} \right). \quad (115)$$

Variation of the curvature scalar: Using (46):

$$\delta^{(1)}R = -\bar{\square}h + \bar{\nabla}^\mu \bar{\nabla}^\nu h_{\mu\nu} - \bar{R}_{\mu\nu} h^{\mu\nu}. \quad (116)$$

Combining all ingredients and performing integrations by parts (discarding surface terms), one arrives at (112).

5.3.3 Compact form with Lichnerowicz

Integrating (112) by parts, the quadratic action can be written compactly as:

$$S_{\text{EH}}^{(2)} = \frac{1}{16\pi G_N} \int d^D x \sqrt{-\bar{g}} \left[\frac{1}{4} h^{\mu\nu} (\Delta_L h)_{\mu\nu} - \frac{1}{4} h \bar{R}_{\mu\nu} h^{\mu\nu} + \text{gauge terms} \right]. \quad (117)$$

In the TT (transverse-traceless) sector, where $\bar{\nabla}^\mu h_{\mu\nu}^{TT} = 0$ and $h^{TT} = 0$, the action reduces to:

$$S_{\text{TT}}^{(2)} = \frac{1}{16\pi G_N} \int d^D x \sqrt{-\bar{g}} \frac{1}{4} h_{\text{TT}}^{\mu\nu} (\Delta_L h^{\text{TT}})_{\mu\nu}. \quad (118)$$

6 Fierz–Pauli theory

6.1 Systematic derivation from the covariant formalism

The *Fierz–Pauli action* is the special case of (112) evaluated on the Minkowski background $\bar{g}_{\mu\nu} = \eta_{\mu\nu}$, where all background curvatures vanish: $\bar{R}_{\mu\nu\rho\sigma} = 0$, $\bar{R}_{\mu\nu} = 0$, $\bar{R} = 0$.

In this limit, the Lichnerowicz operator reduces to:

$$(\Delta_L h)_{\mu\nu} \Big|_\eta = -\square_\eta h_{\mu\nu} \equiv -\eta^{\rho\sigma} \partial_\rho \partial_\sigma h_{\mu\nu}, \quad (119)$$

and the quadratic action (112) becomes:

$$S_{\text{FP}} = \int d^D x \left[-\frac{1}{4} \partial_\rho h_{\mu\nu} \partial^\rho h^{\mu\nu} + \frac{1}{4} \partial_\rho h \partial^\rho h + \frac{1}{2} \partial_\mu h_{\nu\rho} \partial^\nu h^{\mu\rho} - \frac{1}{2} \partial_\mu h \partial_\nu h^{\mu\nu} \right]. \quad (120)$$

6.2 Gauge symmetry of Fierz–Pauli

The action (120) is invariant under the linearized gauge transformation:

$$h_{\mu\nu} \longrightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu, \quad (121)$$

for any vector $\xi_\mu(x)$. This is the linearized version of diffeomorphism invariance.

Verification: Under (121), $\delta h = 2\partial^\mu \xi_\mu$ and $\delta h_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$. Inserting into the action and using integration by parts, all terms cancel.

6.3 Degree-of-freedom count

A symmetric tensor $h_{\mu\nu}$ in D dimensions has $D(D+1)/2$ independent components. The gauge invariance (121) removes D components (the parameters of ξ_μ), and the equations of motion are second order. The D implicit constraint equations (linearized Bianchi identities) further reduce the count. The final result is:

$$\text{physical d.o.f.} = \frac{D(D+1)}{2} - D - D = \frac{(D-2)(D+1)}{2}. \quad (122)$$

In $D = 4$: $\frac{2 \cdot 5}{2} = 5$ components, but two are the non-propagating helicity-0 modes, leaving only **2 propagating modes** — the two polarizations of the massless graviton.

6.4 Fierz–Pauli equations of motion

Varying (120) with respect to $h^{\mu\nu}$:

$$\mathcal{E}_{\mu\nu}[h] = -\frac{1}{2}\Box h_{\mu\nu} - \frac{1}{2}\partial_\mu\partial_\nu h + \partial^\rho\partial_{(\mu}h_{\nu)\rho} - \frac{1}{2}\eta_{\mu\nu}(\partial^\rho\partial^\sigma h_{\rho\sigma} - \Box h) = 0. \quad (123)$$

This is exactly the linearization of the Einstein tensor on a flat background.

6.4.1 Simplification in De Donder gauge

The De Donder (or Lorenz) gauge condition:

$$\partial^\nu h_{\mu\nu} - \frac{1}{2}\partial_\mu h = 0, \quad (124)$$

simplifies the equations of motion to:

$$\Box h_{\mu\nu}^{(\text{DD})} = 0. \quad (125)$$

Additionally, the tracelessness condition $h = 0$ (reachable within De Donder gauge) further simplifies to the standard wave equation for the trace-free perturbation.

6.5 Fierz–Pauli mass term

The massive Fierz–Pauli action adds a unique mass term:

$$S_{\text{FP}}^{(\text{m})} = S_{\text{FP}} - \frac{m^2}{4} \int d^D x (h_{\mu\nu}h^{\mu\nu} - h^2). \quad (126)$$

The term $h_{\mu\nu}h^{\mu\nu} - h^2$ is the unique one that can be added without introducing the *Boulware–Deser ghost* — an extra mode with negative norm. Any other combination $(h_{\mu\nu}h^{\mu\nu} - \alpha h^2)$ with $\alpha \neq 1$ introduces a sixth propagating mode with wrong-sign kinetic term.

The equations of motion of the massive theory are:

$$\Box h_{\mu\nu} - \partial_{(\mu}\partial^\rho h_{\nu)\rho} + \frac{1}{2}\partial_\mu\partial_\nu h - \frac{1}{2}\eta_{\mu\nu}(\Box h - \partial^\rho\partial^\sigma h_{\rho\sigma}) - m^2(h_{\mu\nu} - \eta_{\mu\nu}h) = 0. \quad (127)$$

Taking the divergence of (127) gives the transversality condition $\partial^\mu h_{\mu\nu} = 0$, and taking the trace gives $h = 0$. This fixes all unphysical components, and in the limit $m \rightarrow 0$ the *vDVZ discontinuity* (van Dam–Veltman–Zakharov) occurs.

7 Differential operators on curved backgrounds

7.1 The Lichnerowicz operator: detailed properties

7.1.1 Definition and symmetry

The Lichnerowicz operator (45) acts on symmetric rank-2 tensors. It is convenient to decompose it as:

$$\Delta_L = -\bar{\square} + \hat{E}, \quad (128)$$

where $\hat{E}$ is the endomorphism curvature operator:

$$(\hat{E} h)_{\mu\nu} = -2\bar{R}_{\mu\rho\nu\sigma} h^{\rho\sigma} + \bar{R}^\rho{}_\mu h_{\nu\rho} + \bar{R}^\rho{}_\nu h_{\mu\rho}. \quad (129)$$

The Lichnerowicz operator is self-adjoint with respect to the natural inner product on symmetric tensors:

$$\langle h, \Delta_L k \rangle = \langle \Delta_L h, k \rangle, \quad (130)$$

donde $\langle h, k \rangle = \int d^D x \sqrt{-\bar{g}} h^{\mu\nu} k_{\mu\nu}$.

Proof. The $-\bar{\square}$ part is self-adjoint by integration by parts (without boundary terms, or with appropriate boundary conditions). The $\hat{E}$ part is self-adjoint because the Riemann tensor has the symmetry $R_{\mu\rho\nu\sigma} = R_{\nu\sigma\mu\rho}$. $\square$

7.1.2 On Einstein backgrounds

A background metric is *Einstein* if $\bar{R}_{\mu\nu} = \frac{\bar{R}}{D} \bar{g}_{\mu\nu}$, which in the presence of a cosmological constant implies $\bar{R}_{\mu\nu} = \frac{2\Lambda}{D-2} \bar{g}_{\mu\nu}$.

In this case, the Lichnerowicz operator on TT modes simplifies to:

$$(\Delta_L h^{\text{TT}})_{\mu\nu} \Big|_{\text{Einstein}} = -\bar{\square} h_{\mu\nu}^{\text{TT}} - 2\bar{R}_{\mu\rho\nu\sigma} h^{\text{TT}\rho\sigma} + \frac{2\bar{R}}{D} h_{\mu\nu}^{\text{TT}}. \quad (131)$$

In maximally symmetric backgrounds (AdS_D , dS_D , $\mathbb{M}^D$), the Riemann tensor is:

$$\bar{R}_{\mu\rho\nu\sigma} = \frac{\bar{R}}{D(D-1)} (\bar{g}_{\mu\nu} \bar{g}_{\rho\sigma} - \bar{g}_{\mu\sigma} \bar{g}_{\rho\nu}), \quad (132)$$

and the Lichnerowicz operator on TT modes acts as:

$$(\Delta_L h^{\text{TT}})_{\mu\nu} \Big|_{\text{max. sim.}} = \left(-\bar{\square} + \frac{2\bar{R}}{D(D-1)} \right) h_{\mu\nu}^{\text{TT}}. \quad (133)$$

In maximally symmetric backgrounds, the TT sector of the graviton satisfies:

$$\left(-\bar{\square} + \frac{2\Lambda}{D-2} - m_{\text{eff}}^2 \right) h_{\mu\nu}^{\text{TT}} = 0, \quad (134)$$

where m_{eff}^2 depends on the theory considered. The massless graviton corresponds to $m_{\text{eff}}^2 = 0$ at $\Lambda = 0$.

7.1.3 Spectrum in AdS_D

In anti-de Sitter space AdS_D with radius L ($\Lambda = -(D-1)(D-2)/(2L^2)$), the eigenvalues of the Lichnerowicz operator on TT modes satisfy:

$$-\bar{\square}_{\text{AdS}} \phi = \mu^2 \phi \implies \mu^2 \geq \mu_{\text{BF}}^2, \quad (135)$$

where $\mu_{\text{BF}}^2 = -\frac{(D-1)^2}{4L^2}$ is the *Breitenlöhner–Freedman (BF) bound*. Modes with $\mu_{\text{BF}}^2 \leq \mu^2 < 0$ are stable in AdS but would be unstable in flat space — a phenomenon of stabilization by negative curvature.

7.2 The Dirac operator on curved backgrounds

7.2.1 Definition

To analyze fermions on curved backgrounds, we need the *vielbein* e^a_μ such that $g_{\mu\nu} = e^a_\mu e^b_\nu \eta_{ab}$, and the *spin connection*:

$$\omega_\mu^{ab} = e^{\nu a} (\partial_\mu e^b_\nu - \Gamma_{\mu\nu}^\rho e^b_\rho). \quad (136)$$

The covariant derivative on spinors is:

$$\nabla_\mu \psi = \partial_\mu \psi + \frac{1}{4} \omega_\mu^{ab} \gamma_{ab} \psi, \quad \gamma_{ab} = \frac{1}{2} [\gamma_a, \gamma_b]. \quad (137)$$

The *Dirac operator* on the curved background is:

$$D = \gamma^\mu \nabla_\mu = e_a^\mu \gamma^a \nabla_\mu. \quad (138)$$

7.2.2 Lichnerowicz–Weitzenböck identity

The square of the Dirac operator satisfies the fundamental identity:

$$D^2 = -\square_{\text{spinor}} + \frac{1}{4} R, \quad (139)$$

where $\square_{\text{spinor}} = g^{\mu\nu} \nabla_\mu \nabla_\nu$ acts on spinors, and R is the background curvature scalar.

Proof. We compute $D^2 = \gamma^\mu \nabla_\mu \gamma^\nu \nabla_\nu$. Using $\gamma^\mu \gamma^\nu = g^{\mu\nu} + \frac{1}{2} [\gamma^\mu, \gamma^\nu]$:

$$\begin{aligned} D^2 &= g^{\mu\nu} \nabla_\mu \nabla_\nu + \frac{1}{2} \gamma^{\mu\nu} [\nabla_\mu, \nabla_\nu] \\ &= -\square + \frac{1}{4} \gamma^{\mu\nu} \gamma^{\rho\sigma} R_{\mu\nu\rho\sigma}, \end{aligned} \quad (140)$$

where $\gamma^{\mu\nu} = \frac{1}{2} [\gamma^\mu, \gamma^\nu]$. The cyclic Bianchi identity and antisymmetry of the Riemann tensor imply $\gamma^{\mu\nu} \gamma^{\rho\sigma} R_{\mu\nu\rho\sigma} = R$, giving (139). $\square$

7.2.3 Physical consequences

The identity (139) has deep consequences:

- (i) If $R > 0$ everywhere (as in dS or spheres), $D^2 \geq \frac{R}{4} > 0$, and the Dirac operator has no zero modes (no nontrivial harmonic spinors). This implies the Atiyah–Singer index vanishes.

- (ii) On Ricci-flat backgrounds ($R_{\mu\nu} = 0$), $D^2 = -\square_{\text{spinor}}$, and the spectrum of the Dirac operator directly determines that of the spinor Laplacian.
- (iii) The relation (139) is the basis of *positive energy proofs* in general relativity (Witten, 1981).

7.3 Relations between operators

Table 1 summarizes the relations between the relevant operators.

Table 1: Relations between second-order differential operators on curved backgrounds.

Operator	Acts on	Expression
Δ_L	Symmetric tensors	$-\bar{\square} - 2\hat{R} + \hat{\text{Ric}}$
D^2	Espinors	$-\square + \frac{1}{4}R$
Δ_{Hodge}	p -formas	$-(d\delta + \delta d)$
$\square_{\text{Lichnerowicz}}$	Funciones	$-\bar{\nabla}^\mu \bar{\nabla}_\mu$

The key relation between Δ_L and D^2 in supersymmetric backgrounds allows *transferring* spectral information between the bosonic and fermionic sectors, which is central to the stability proof in supersymmetric compactifications.

8 Tensor harmonics and spectral decomposition

8.1 York decomposition

Any symmetric tensor $h_{\mu\nu}$ on an appropriate spacetime admits the *York decomposition*:

$$h_{\mu\nu} = h_{\mu\nu}^{\text{TT}} + \bar{\nabla}_\mu V_\nu + \bar{\nabla}_\nu V_\mu + \bar{\nabla}_\mu \bar{\nabla}_\nu \sigma - \frac{1}{D} \bar{g}_{\mu\nu} \bar{\square} \sigma + \frac{1}{D} \bar{g}_{\mu\nu} \phi, \quad (141)$$

where:

- $h_{\mu\nu}^{\text{TT}}$: transverse-traceless tensor ($\bar{\nabla}^\mu h_{\mu\nu}^{\text{TT}} = 0$, $\bar{g}^{\mu\nu} h_{\mu\nu}^{\text{TT}} = 0$),
- V_μ : transverse vector ($\bar{\nabla}^\mu V_\mu = 0$),
- σ, ϕ : scalars.

This decomposition is orthogonal with respect to the inner product $\langle h, k \rangle = \int \sqrt{-\bar{g}} h^{\mu\nu} k_{\mu\nu}$.

8.2 Tensor harmonics on maximally symmetric backgrounds

On maximally symmetric backgrounds (sphere S^n , AdS_D , dS_D), the tensor harmonics $Y_{\mu\nu}^{(\ell)}$ are simultaneous eigenfunctions of the Lichnerowicz operator and the Casimir operators of the isometry group:

$$\Delta_L Y_{\mu\nu}^{(\ell)} = \lambda_\ell^{\text{TT}} Y_{\mu\nu}^{(\ell)}, \quad (142)$$

with eigenvalues

$$\lambda_\ell^{\text{TT}} = \frac{\ell(\ell + D - 1) - 2}{L^2} \quad (\text{en } S^D \text{ de radio } L). \quad (143)$$

The vector harmonics $Y_\mu^{(\ell)}$ and scalar harmonics $Y^{(\ell)}$ satisfy analogous equations, and the entire spectrum is discrete and bounded from below.

8.3 Mode expansion and quantization

The perturbation is expanded as:

$$h_{\mu\nu}(x) = \sum_n a_n Y_{\mu\nu}^{(n)}(x), \quad (144)$$

where the coefficients a_n are the physical degrees of freedom of the graviton. Upon quantization, $a_n \rightarrow \hat{a}_n$ become creation and annihilation operators with standard bosonic algebra.

The oscillation frequency of each mode is determined by the eigenvalue:

$$\omega_n^2 = \lambda_n + m_{\text{eff}}^2. \quad (145)$$

The stability condition is $\omega_n^2 > 0$ for all n .

9 Applications to stability

9.1 Linear stability: general criterion

Given a background $\bar{g}_{\mu\nu}$ and a quadratic action $S^{(2)}[h]$, the linear stability criterion is:

The background $\bar{g}_{\mu\nu}$ is *linearly stable* if and only if the operator $\mathcal{O}$ in $S^{(2)} = \int h \mathcal{O} h$ satisfies:

$$\text{spec}(\mathcal{O}) \Big|_{\text{physical}} > 0. \quad (146)$$

A mode with $\lambda < 0$ signals a tachyonic instability; a mode with $\lambda = 0$ signals a flat direction (zero mode, possibly associated with a modulus of the solution space).

9.2 Black hole stability

For static spherically symmetric black holes, TT tensor perturbations satisfy:

$$\left[-\partial_t^2 + \partial_{r_*}^2 - V_\ell(r) \right] \Psi_\ell = 0, \quad (147)$$

where r_* is the tortoise coordinate and $V_\ell(r)$ is the *Regge–Wheeler potential* (for axial perturbations) or the *Zerilli potential* (for polar perturbations), which in the limit $\ell \gg 1$ is governed by the Lichnerowicz operator eigenvalue:

$$V_\ell(r) \sim \frac{\lambda_\ell^{\text{TT}}}{r^2} f(r) + \mathcal{O}(r^{-3}), \quad f(r) = 1 - \frac{r_s}{r}. \quad (148)$$

The Gregory–Laflamme instability of black strings is a paradigmatic example of a mode with $\omega^2 < 0$.

9.3 $f(R)$ theories and additional degrees of freedom

For the action $S = \int \sqrt{-g} f(R)$, the quadratic action takes the form:

$$S_{f(R)}^{(2)} = \int d^D x \sqrt{-\bar{g}} \left[f'(\bar{R}) (\text{EH terms}) + f''(\bar{R}) (\delta R)^2 \right]. \quad (149)$$

The term $f''(\bar{R})(\delta R)^2$ introduces an *additional scalar* with mass:

$$m_{\text{escalar}}^2 = \frac{f'(\bar{R})}{3f''(\bar{R})} - \frac{2\Lambda}{D-2}. \quad (150)$$

The condition $m_{\text{scalar}}^2 > 0$ is necessary for the absence of scalar instabilities, and $f''(\bar{R}) > 0$ is necessary for the absence of ghosts.

9.4 Ostrogradsky instabilities

In theories with higher-order derivatives, the kinetic operator is not Δ_L but a fourth-order operator $\mathcal{O}^{(4)}$. The spectrum of $\mathcal{O}^{(4)}$ may include *negative-norm states* (ghosts), leading to quantum instabilities. The Ostrogradsky criterion states that theories with more than two derivatives (in the reduced action) inevitably generate such ghosts, unless the theory is degenerate in the appropriate sense (DHOST theories).

10 Final remarks and outlook

These notes have presented the linearization formalism of gravity with emphasis on the central role of the Lichnerowicz operator. Topics still to be developed include:

- (i) **Explicit computation of the Δ_L spectrum in Schwarzschild–AdS:** The eigenvalues of the Lichnerowicz operator in this background determine the quasi-normal mode frequencies of the black hole.
- (ii) **Relation to the AdS/CFT correspondence:** The graviton TT modes in AdS correspond to the stress tensor of the dual field theory.
- (iii) **Extension to supergravity:** In this context, the Lichnerowicz and Dirac operators are related by supersymmetry, and the stability of the bosonic background is guaranteed by the positivity of the Dirac operator.
- (iv) **Massive gravity theories:** The extension of Fierz–Pauli to the nonlinear regime (de Rham–Gabadadze–Tolley gravity, dRGT) and the ultraviolet completion problem.

A Useful tensor identities

A.1 Bianchi identities

First Bianchi identity (algebraic):

$$R_{\mu[\nu\rho\sigma]} = 0 \quad \Longleftrightarrow \quad R_{\mu\nu\rho\sigma} + R_{\mu\rho\sigma\nu} + R_{\mu\sigma\nu\rho} = 0. \quad (151)$$

Second (differential) Bianchi identity:

$$\nabla_{[\lambda} R_{\mu\nu]\rho\sigma} = 0, \quad (152)$$

which implies the contracted identity:

$$\nabla^\mu G_{\mu\nu} = 0, \quad G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R. \quad (153)$$

A.2 Commutators of covariant derivatives

For a vector V^μ :

$$[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho_{\sigma\mu\nu}V^\sigma. \quad (154)$$

For a rank-2 tensor:

$$[\nabla_\rho, \nabla_\sigma]h_{\mu\nu} = -R^\lambda_{\mu\rho\sigma}h_{\lambda\nu} - R^\lambda_{\nu\rho\sigma}h_{\mu\lambda}. \quad (155)$$

A.3 Variation of geometric quantities

$$\delta\Gamma^\rho_{\mu\nu} = \frac{1}{2}g^{\rho\sigma}(\nabla_\mu\delta g_{\nu\sigma} + \nabla_\nu\delta g_{\mu\sigma} - \nabla_\sigma\delta g_{\mu\nu}), \quad (156)$$

$$\delta R_{\mu\nu} = \nabla^\rho\delta\Gamma_{\rho(\mu\nu)} - \nabla_{(\mu}\delta\Gamma^\rho_{|\rho|\nu)}, \quad (157)$$

$$\delta R = -R^{\mu\nu}\delta g_{\mu\nu} + g^{\mu\nu}\delta R_{\mu\nu}, \quad (158)$$

$$\delta\sqrt{-g} = \frac{1}{2}\sqrt{-g}g^{\mu\nu}\delta g_{\mu\nu}. \quad (159)$$

B SVT decomposition in perturbed cosmology

On an FRW background with metric $ds^2 = -dt^2 + a^2(t)\gamma_{ij}dx^i dx^j$, the metric perturbation decomposes into *scalar-vector-tensor* (SVT) modes:

$$h_{00} = -2\Phi, \quad (160)$$

$$h_{0i} = a(B_{,i} + S_i), \quad (161)$$

$$h_{ij} = a^2(-2\Psi\gamma_{ij} + 2E_{,ij} + F_{i,j} + F_{j,i} + h_{ij}^{TT}), \quad (162)$$

where Φ, Ψ, B, E are scalars; S_i, F_i are transverse vectors; and h_{ij}^{TT} is the transverse-traceless tensor (gravitational waves).

In the gauge $B = E = 0$ (synchronous gauge) or $\Phi = B$ (longitudinal or Newtonian gauge), the perturbation equations decouple into independent SVT sectors at linear order, greatly simplifying the analysis.

C De Donder gauge: existence and uniqueness

Given any perturbation $h_{\mu\nu}$, there exists a diffeomorphism generated by ξ_μ such that the transformed perturbation $h'_{\mu\nu} = h_{\mu\nu} + \bar{\nabla}_\mu\xi_\nu + \bar{\nabla}_\nu\xi_\mu$ satisfies the De Donder condition:

$$\bar{\nabla}^\nu h'_{\mu\nu} - \frac{1}{2}\bar{\nabla}_\mu h' = 0. \quad (163)$$

Proof. The De Donder condition translates into a differential equation for ξ_μ :

$$\bar{\square}\xi_\mu + \bar{R}^\nu{}_\mu \xi_\nu = -\left(\bar{\nabla}^\nu h_{\mu\nu} - \frac{1}{2}\bar{\nabla}_\mu h\right). \quad (164)$$

The operator on the left-hand side is the Hodge Laplacian acting on 1-forms, which on backgrounds without zero modes (a condition satisfied on strictly hyperbolic backgrounds) has a well-defined inverse. $\square$

The residual gauge freedom within De Donder gauge corresponds to homogeneous solutions $\bar{\square}\xi_\mu + \bar{R}^\nu{}_\mu \xi_\nu = 0$, which in flat space are simply plane waves.

D Explicit derivation of the Lichnerowicz operator from $\delta R_{\mu\nu}$

This section develops in detail how the operator Δ_L emerges when reorganizing the variation of the Ricci tensor, identifying each term precisely. It bridges the notation of the TFM of Fernández Uria [1] y la forma compacta usada en las Secciones anteriores de estas notas.

D.1 Starting point: variation of the Ricci tensor

Recall the exact expression (Section 2.5):

$$\delta R_{\mu\nu} = -\frac{1}{2}\bar{\square}h_{\mu\nu} - \frac{1}{2}\bar{\nabla}_\mu \bar{\nabla}_\nu h + \bar{\nabla}^\rho \bar{\nabla}_{(\mu} h_{\nu)\rho} - \bar{R}_{\mu\rho\nu\sigma} h^{\rho\sigma} + \bar{R}^\rho{}_{(\mu} h_{\nu)\rho}. \quad (165)$$

We want to rewrite this in terms of the Lichnerowicz operator and a “gauge term”.

D.2 Identification of the Lichnerowicz operator

We define the *trace-reversed divergence* of the perturbation:

$$C_\mu \equiv \bar{\nabla}^\nu h_{\mu\nu} - \frac{1}{2}\bar{\nabla}_\mu h. \quad (166)$$

This is the De Donder gauge condition when $C_\mu = 0$.

The variation of the Ricci tensor decomposes as:

$$\delta R_{\mu\nu} = -\frac{1}{2}(\Delta_L h)_{\mu\nu} + \bar{\nabla}_{(\mu} C_{\nu)}, \quad (167)$$

donde Δ_L es el operador de Lichnerowicz definido en (45).

Proof. We compute $\bar{\nabla}_{(\mu} C_{\nu)}$ explicitly:

$$\begin{aligned} \bar{\nabla}_{(\mu} C_{\nu)} &= \frac{1}{2}(\bar{\nabla}_\mu C_\nu + \bar{\nabla}_\nu C_\mu) \\ &= \frac{1}{2}(\bar{\nabla}_\mu \bar{\nabla}^\rho h_{\nu\rho} - \frac{1}{2}\bar{\nabla}_\mu \bar{\nabla}_\nu h + \bar{\nabla}_\nu \bar{\nabla}^\rho h_{\mu\rho} - \frac{1}{2}\bar{\nabla}_\nu \bar{\nabla}_\mu h). \end{aligned} \quad (168)$$

We now subtract (168) from (165):

$$\begin{aligned}\delta R_{\mu\nu} - \bar{\nabla}_{(\mu} C_{\nu)} &= -\frac{1}{2}\bar{\square}h_{\mu\nu} - \frac{1}{2}\bar{\nabla}_{\mu}\bar{\nabla}_{\nu}h + \bar{\nabla}^{\rho}\bar{\nabla}_{(\mu}h_{\nu)\rho} - \bar{R}_{\mu\rho\nu\sigma}h^{\rho\sigma} + \bar{R}^{\rho}_{(\mu}h_{\nu)\rho} \\ &\quad - \frac{1}{2}\bar{\nabla}_{(\mu}\bar{\nabla}^{\rho}h_{\nu)\rho} + \frac{1}{4}\bar{\nabla}_{\mu}\bar{\nabla}_{\nu}h + \frac{1}{4}\bar{\nabla}_{\nu}\bar{\nabla}_{\mu}h \\ &= -\frac{1}{2}\bar{\square}h_{\mu\nu} + \frac{1}{2}\bar{\nabla}^{\rho}\bar{\nabla}_{(\mu}h_{\nu)\rho} - \bar{R}_{\mu\rho\nu\sigma}h^{\rho\sigma} + \bar{R}^{\rho}_{(\mu}h_{\nu)\rho}.\end{aligned}\quad (169)$$

We need to simplify the term $\frac{1}{2}\bar{\nabla}^{\rho}\bar{\nabla}_{(\mu}h_{\nu)\rho}$. Using the commutator identity of covariant derivatives on symmetric tensors:

$$\bar{\nabla}^{\rho}\bar{\nabla}_{\mu}h_{\nu\rho} = \bar{\nabla}_{\mu}\bar{\nabla}^{\rho}h_{\nu\rho} + \bar{R}^{\rho}_{\mu}h_{\nu\rho} - \bar{R}^{\rho\sigma}_{\mu\nu}h_{\rho\sigma}, \quad (170)$$

so that:

$$\frac{1}{2}\bar{\nabla}^{\rho}\bar{\nabla}_{(\mu}h_{\nu)\rho} = \frac{1}{2}\left(\bar{\nabla}_{\mu}\bar{\nabla}^{\rho}h_{\nu\rho} + \bar{\nabla}_{\nu}\bar{\nabla}^{\rho}h_{\mu\rho}\right) \underbrace{- \bar{R}_{\mu\rho\nu\sigma}h^{\rho\sigma} + \bar{R}^{\rho}_{(\mu}h_{\nu)\rho}}_{\text{curvatura}}. \quad (171)$$

Substituting into (169):

$$\delta R_{\mu\nu} - \bar{\nabla}_{(\mu} C_{\nu)} = -\frac{1}{2}\bar{\square}h_{\mu\nu} - 2\bar{R}_{\mu\rho\nu\sigma}h^{\rho\sigma} + 2\bar{R}^{\rho}_{(\mu}h_{\nu)\rho}. \quad (172)$$

We recognize exactly $-\frac{1}{2}(\Delta_L h)_{\mu\nu}$, which proves (167). $\square$

D.3 Consequence: equations of motion in De Donder gauge

In De Donder gauge ($C_{\mu} = 0$), the variation of the Ricci tensor is directly:

$$\delta R_{\mu\nu}\Big|_{C=0} = -\frac{1}{2}(\Delta_L h)_{\mu\nu}. \quad (173)$$

The linearized Einstein equations $\delta G_{\mu\nu} = -\kappa t_{\mu\nu}$ (in vacuum, $t_{\mu\nu} = 0$) then reduce, using (173) and the corresponding trace, to:

$$(\Delta_L h)_{\mu\nu} - \frac{1}{2}\bar{g}_{\mu\nu}\bar{g}^{\rho\sigma}(\Delta_L h)_{\rho\sigma} + \bar{R}_{\mu\nu}{}^{\rho\sigma}h_{\rho\sigma}[\text{background terms}] = 0. \quad (174)$$

In the *TT sector* (transverse-traceless), $C_{\mu} = 0$ and $h = 0$, so the equation becomes simply:

$$\boxed{(\Delta_L h^{\text{TT}})_{\mu\nu} = 0.} \quad (175)$$

This is the master equation of the graviton on curved backgrounds.

E Linearized Bianchi identity and perturbed conservation

E.1 The Bianchi identity at first order

The contracted Bianchi identity $\bar{\nabla}^{\mu}G_{\mu\nu} = 0$ persists when the metric is perturbed. The explicit form of its linearization reveals the automatic consistency of the equations of motion.

The variation of the Einstein tensor satisfies:

$$\bar{\nabla}^\mu(\delta G_{\mu\nu}) + (\delta\Gamma^\mu_{\mu\lambda})\bar{G}^\lambda_{\nu} - (\delta\Gamma^\lambda_{\mu\nu})\bar{G}^\mu_{\lambda} = 0, \quad (176)$$

which on the vacuum background ($\bar{G}_{\mu\nu} = 0$) reduces to $\bar{\nabla}^\mu(\delta G_{\mu\nu}) = 0$.

Esquema. We start from $\nabla^\mu G_{\mu\nu} = 0$ and vary with respect to the perturbation $h_{\rho\sigma}$. The Leibniz rule gives:

$$\delta(\nabla^\mu G_{\mu\nu}) = (\delta g^{\mu\lambda})\bar{\nabla}_\lambda \bar{G}_{\mu\nu} + \bar{g}^{\mu\lambda}[\delta(\nabla_\lambda G_{\mu\nu})] = 0. \quad (177)$$

Expanding the perturbed covariant derivative:

$$\delta(\nabla_\lambda G_{\mu\nu}) = \bar{\nabla}_\lambda(\delta G_{\mu\nu}) + (\delta\Gamma^\rho_{\lambda\mu})G_{\rho\nu} + (\delta\Gamma^\rho_{\lambda\nu})G_{\mu\rho} - (\delta\Gamma^\rho_{\lambda\rho})G_{\mu\nu}. \quad (178)$$

Contracting with $\bar{g}^{\mu\lambda}$ and using the antisymmetry of the perturbed Christoffel symbols, together with the symmetry of $\bar{G}_{\mu\nu}$, one arrives at (176). $\square$

E.2 Implication: conservation of the perturbed energy-momentum tensor

In the presence of matter, the linearized Einstein equation is $\delta G_{\mu\nu} = -\kappa t_{\mu\nu}$. The linearized Bianchi identity then implies:

$$\bar{\nabla}^\mu t_{\mu\nu} = 0, \quad (179)$$

on the vacuum background. This is the *covariant conservation* of the perturbed energy-momentum tensor, guaranteeing the consistency of the problem: the linearized Einstein equations are automatically compatible with matter conservation.

Remark on equation counting. The tensor $\delta G_{\mu\nu}$ has $D(D+1)/2$ components, but the D linearized Bianchi identities (176) imply that only $D(D+1)/2 - D = D(D-1)/2$ are independent. This is consistent with the physical degrees of freedom of the graviton.

E.3 Explicit verification on a flat background

On the Minkowski background, $\bar{G}_{\mu\nu} = 0$ and $\bar{\nabla}_\mu \rightarrow \partial_\mu$. The linearized Bianchi identity takes the form:

$$\partial^\mu(\delta G_{\mu\nu}) = 0, \quad (180)$$

which can be verified directly from the Fierz-Pauli equation (123):

$$\partial^\mu \mathcal{E}_{\mu\nu} = -\frac{1}{2}\partial^\mu \square h_{\mu\nu} + \frac{1}{2}\partial^\mu \partial_\nu \partial^\rho h_{\mu\rho} - \frac{1}{2}\partial^\mu \partial_\mu \partial^\rho h_{\nu\rho} + \frac{1}{4}\partial_\nu \square h - \frac{1}{4}\partial_\nu \square h = 0. \quad (181)$$

All terms cancel due to the symmetry of partial derivatives, confirming consistency.

F De Donder gauge on curved backgrounds: complete analysis

F.1 Motivation: structure of the EOM

The linearized Einstein equations (48) contain terms of the form $\bar{\nabla}_{(\mu} C_{\nu)}$, where C_μ is the “gauge divergence” (166). These contributions vanish in De Donder gauge, greatly simplifying the analysis.

F.2 Explicit form of the condition

The De Donder condition is:

$$C_\mu \equiv \bar{\nabla}^\nu h_{\mu\nu} - \frac{1}{2} \bar{\nabla}_\mu h = 0. \quad (182)$$

In terms of the trace-reversed perturbation $\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2} \bar{g}_{\mu\nu} h$ (where $\bar{h} = -h$ in $D = 4$, and $\bar{h} = -\frac{D-2}{2}h$ in D dimensions), the condition (182) is equivalent to:

$$\bar{\nabla}^\nu \bar{h}_{\mu\nu} = 0. \quad (183)$$

This is the *gravitational Lorenz condition*, analogous to the Lorenz gauge in electromagnetism ($\partial^\mu A_\mu = 0$).

F.3 Existence via the vector wave operator

To prove that De Donder gauge is always reachable, we need to show that the equation for the generating vector ξ_μ has a solution.

Under the gauge transformation $h_{\mu\nu} \rightarrow h_{\mu\nu} + 2\bar{\nabla}_{(\mu} \xi_{\nu)}$, the gauge divergence transforms as:

$$C_\mu \rightarrow C_\mu + \mathcal{L}_\xi C_\mu^{\text{gauge}}, \quad (184)$$

where the gauge term is:

$$\begin{aligned} \delta_\xi C_\mu &= \bar{\nabla}^\nu (2\bar{\nabla}_{(\mu} \xi_{\nu)}) - \frac{1}{2} \bar{\nabla}_\mu (2\bar{\nabla}^\nu \xi_\nu) \\ &= \bar{\square} \xi_\mu + \bar{\nabla}_\mu (\bar{\nabla}^\nu \xi_\nu) - \bar{\nabla}_\mu (\bar{\nabla}^\nu \xi_\nu) + \bar{R}^\nu{}_\mu \xi_\nu \\ &= \bar{\square} \xi_\mu + \bar{R}^\nu{}_\mu \xi_\nu \equiv \mathcal{D}_\mu [\xi], \end{aligned} \quad (185)$$

where we have used the commutator identity $[\bar{\nabla}^\nu, \bar{\nabla}_\mu] \xi_\nu = \bar{R}^\nu{}_\mu \xi_\nu$.

The operator $\mathcal{D}_\mu = \bar{\square} + \hat{\bar{R}}$ is the *Lichnerowicz vector wave operator* (acting on 1-forms), and is the Hodge Laplacian on vectors:

$$\mathcal{D}_\mu [\xi] = -(dd^\dagger + d^\dagger d) \xi \Big|_\mu = (\Delta_{\text{Hodge}} \xi)_\mu. \quad (186)$$

To reach $C'_\mu = 0$ starting from C_μ , we need to solve:

$$\bar{\square} \xi_\mu + \bar{R}^\nu{}_\mu \xi_\nu = -C_\mu. \quad (187)$$

This is a wave equation (in Lorentzian spacetime) or an elliptic equation (in Riemannian signature) for the vector ξ_μ .

On globally hyperbolic Lorentzian backgrounds (with appropriate initial data on a Cauchy hypersurface), equation (187) has a unique solution ξ_μ for any given C_μ . Therefore, De Donder gauge is always reachable.

The existence of solutions follows from the existence theorem for hyperbolic wave operators on globally hyperbolic manifolds (Leray's theorem).

F.4 Residual gauge freedom

Once in De Donder gauge, further transformations can be made with vectors ξ_μ satisfying the homogeneous equation:

$$\bar{\square}\xi_\mu + \bar{R}^\nu{}_\mu \xi_\nu = 0. \quad (188)$$

This freedom can be used to impose additional conditions on $h_{\mu\nu}$, such as the tracelessness condition $h = 0$, leading to the *TT gauge* (transverse-traceless) in the tensor sector.

Full TT gauge. In the tensor sector of perturbations (without scalars or vectors), we can simultaneously impose:

$$\bar{\nabla}^\nu h_{\mu\nu}^{\text{TT}} = 0, \quad h^{\text{TT}} = 0. \quad (189)$$

These $D + 1$ conditions eliminate all gauge degrees of freedom, leaving only the physical propagating modes.

F.5 EOM in De Donder gauge: compact form

In De Donder gauge (182), the linearized Einstein equations take the form:

$$-\frac{1}{2}(\Delta_L h)_{\mu\nu} - \frac{1}{2}\bar{g}_{\mu\nu} \left[\bar{\nabla}^\rho \bar{\nabla}^\sigma h_{\rho\sigma} - \bar{\square}h - \bar{R}^{\rho\sigma} h_{\rho\sigma} \right] = \kappa t_{\mu\nu}. \quad (190)$$

Using $C_\mu = 0 \Rightarrow \bar{\nabla}^\nu h_{\mu\nu} = \frac{1}{2}\bar{\nabla}_\mu h$, the trace terms simplify further. In the TT sector (189):

$$\boxed{(\Delta_L h^{\text{TT}})_{\mu\nu} = 2\kappa t_{\mu\nu}^{\text{TT}}.} \quad (191)$$

In vacuum ($t_{\mu\nu} = 0$), the TT graviton satisfies the covariant wave equation $\Delta_L h^{\text{TT}} = 0$, whose solutions are the physical gravitational waves.

G Explicit SVT decomposition and operators in each sector

G.1 York decomposition in four dimensions

In $D = 4$, the York decomposition (141) has the following degrees of freedom, counting independent components:

Mode	Symbol	d.o.f.	Sector
TT tensor	h_{ij}^{TT}	2	tensor
Transverse vector	V_i	2	vector
Scalar 1	ϕ	1	scalar
Scalar 2	σ	1	scalar
Total	$h_{\mu\nu}$	6	—

Note: in $D = 4$ there are $4 \times 5/2 = 10$ components of $h_{\mu\nu}$, of which 4 are gauge (ξ_μ) , leaving 6 physical. The 2 TT modes are the only ones propagating in vacuum.

G.2 The Lichnerowicz operator in each sector

The key feature of the formalism is that, on maximally symmetric or Einstein backgrounds, the Lichnerowicz operator acts simply in each sector:

TT tensor sector. On Einstein backgrounds with $\bar{R}_{\mu\nu} = \Lambda_{\text{eff}} \bar{g}_{\mu\nu}$:

$$(\Delta_L h^{\text{TT}})_{ij} = \left(-\bar{\square} - \frac{2\bar{R}}{D(D-1)} \right) h_{ij}^{\text{TT}} + 2\bar{R}_{ikjl} h^{\text{TT}kl}, \quad (192)$$

which on maximally symmetric backgrounds (where $\bar{R}_{ijkl}$ takes the standard form) reduces to a scalar operator acting on h_{ij}^{TT} .

Vector sector. The transverse vector V_i satisfies the wave equation for effective massive vector fields, where the effective mass comes from the background curvature.

Scalar sector. The scalars ϕ and σ mix in general and satisfy a coupled system of equations, whose decoupling requires De Donder gauge and trace conditions.

G.3 Kinetic operators in each sector

We summarize the kinetic operators $\mathcal{K}$ appearing in $S^{(2)} = \int \Psi \cdot \mathcal{K} \Psi$ for each sector:

$$\mathcal{K}_{ijkl}^{\text{TT}} = \frac{1}{2} (\Delta_L)_{ijkl}, \quad (193)$$

$$\mathcal{K}_{ij}^{\text{vec}} = \bar{\square} \delta_{ij} + \bar{R}_{ij}, \quad (194)$$

$$\mathcal{K}^{\text{scal}} = \begin{pmatrix} \bar{\square} - \frac{D-1}{D-2} \bar{R} & -\frac{1}{D-2} \bar{\nabla}^2 \\ -\frac{1}{D-2} \bar{\nabla}^2 & \bar{\square} \end{pmatrix}, \quad (195)$$

where $\bar{\nabla}^2 = \bar{g}^{ij} \bar{\nabla}_i \bar{\nabla}_j$ is the spatial Laplacian. Stability of the background requires all eigenvalues of these operators to be non-negative.

H Regge–Wheeler and Zerilli potentials: derivation and spectrum

This section develops in detail the perturbation equation of Schwarzschild black holes, the Regge–Wheeler and Zerilli potentials, and the spectrum of quasinormal modes (QNMs).

H.1 Schwarzschild metric and perturbations

The Schwarzschild metric in standard coordinates is:

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2d\Omega_2^2, \quad f(r) = 1 - \frac{2M}{r}, \quad (196)$$

where $d\Omega_2^2 = d\theta^2 + \sin^2\theta d\phi^2$ is the unit 2-sphere metric.

The Schwarzschild spacetime has isometry group $\mathbb{R} \times SO(3)$, which allows decomposing perturbations into tensor spherical harmonics $Y^{(\ell,m)}(\theta, \phi)$ with $\ell = 0, 1, 2, \dots$ and $|m| \leq \ell$.

H.2 Classification of perturbations

Under parity (reflection $(\theta, \phi) \rightarrow (\pi - \theta, \pi + \phi)$), the tensor harmonics are classified as:

Axial parity (odd parity, “axial” or Regge–Wheeler type). They transform as $(-1)^{\ell+1}$ under parity. The relevant axial perturbations are:

$$h_{\mu\nu}^{\text{axial}} dx^\mu dx^\nu = \sum_{\ell,m} \left[h_0(t, r) Y^{\ell m}(\sin\theta \partial_\phi) dt + h_1(t, r) Y^{\ell m}(\sin\theta \partial_\phi) dr \right]. \quad (197)$$

Polar parity (even parity, “polar” or Zerilli type). They transform as $(-1)^\ell$ under parity and include the metric perturbations H_0, H_1, H_2, K plus perturbations of the lapse function.

The two sectors decouple at linear level due to the parity symmetry of the background metric.

H.3 Regge–Wheeler equation (axial sector)

For the axial sector with $\ell \geq 2$, we introduce the Regge–Wheeler function $\Psi_\ell^{(-)}(t, r)$ defined by:

$$\Psi_\ell^{(-)} = \frac{f(r)}{r} h_1(t, r). \quad (198)$$

Substituting into the linearized Einstein equations and separating into modes $e^{-i\omega t}$, one obtains the *Regge–Wheeler equation*:

$$\boxed{\frac{d^2 \Psi_\ell^{(-)}}{dr_*^2} + [\omega^2 - V_\ell^{(-)}(r)] \Psi_\ell^{(-)} = 0,} \quad (199)$$

where the *tortoise coordinate* r_* is defined by:

$$\frac{dr_*}{dr} = \frac{1}{f(r)}, \quad r_* = r + 2M \ln \left| \frac{r}{2M} - 1 \right|, \quad (200)$$

and the *Regge–Wheeler potential* is:

$$V_\ell^{(-)}(r) = f(r) \left[\frac{\ell(\ell+1)}{r^2} - \frac{6M}{r^3} \right]. \quad (201)$$

Interpretation. Equation (199) is a one-dimensional stationary Schrödinger equation in the “potential” $V_\ell^{(-)}(r)$, with $r_* \in (-\infty, +\infty)$. The event horizon $r = 2M$ corresponds to $r_* \rightarrow -\infty$ and spatial infinity $r \rightarrow \infty$ to $r_* \rightarrow +\infty$.

H.4 El potencial de Regge–Wheeler: propiedades

- (i) **Asymptotic limits:** $V_\ell^{(-)} \rightarrow 0$ tanto para $r_* \rightarrow -\infty$ (horizonte) como para $r_* \rightarrow +\infty$ (infinito espacial).
- (ii) **Positividad:** Para $\ell \geq 2$, el potencial es *no negativo en todo el exterior del agujero negro* ($r > 2M$), lo que implica la estabilidad lineal del sector axial.
- (iii) **Maximum:** The potential has a unique maximum near $r \approx 3M$ (the photosphere of the black hole), with approximate value:

$$V_{\ell, \max}^{(-)} \approx \frac{\ell(\ell+1) - 2}{27M^2} \quad (\text{para } \ell \gg 1). \quad (202)$$

- (iv) **Positividad del espectro:** La positividad de $V_\ell^{(-)}$ implica que no existen modos con $\omega^2 < 0$ (modos inestables) en el sector axial. La prueba rigurosa usa el spectral analysis of the resulting Schrödinger operator.

H.5 Zerilli equation (polar sector)

For the polar sector with $\ell \geq 2$, the analysis is more involved due to the mixing of the fields H_0, H_1, H_2, K . Zerilli [4] showed that via an appropriate linear combination of these fields one can define a single master function $\Psi_\ell^{(+)}(t, r)$ satisfying:

$$\boxed{\frac{d^2 \Psi_\ell^{(+)}}{dr_*^2} + [\omega^2 - V_\ell^{(+)}(r)] \Psi_\ell^{(+)} = 0,} \quad (203)$$

with the *Zerilli potential*:

$$V_\ell^{(+)}(r) = f(r) \cdot \frac{2\Lambda^2 [\Lambda^2 r^2 + 3\Lambda M r + 9M^2] + 18M^2(r - 2M)[\Lambda r + 3M]/r^3}{r^3(\Lambda r + 3M)^2}, \quad (204)$$

where we have defined:

$$\Lambda \equiv \frac{1}{2}(\ell - 1)(\ell + 2) = \frac{1}{2}(\ell^2 + \ell - 2). \quad (205)$$

Comparison with the axial potential. For $\ell = 2$: $\Lambda = 2$, and the difference between $V_2^{(+)}$ and $V_2^{(-)}$ is significant near the horizon, but both share the same asymptotic limits and the same global positivity.

H.6 Isospectrality of Regge–Wheeler and Zerilli

A remarkable result, discovered by Chandrasekhar [5], is that the two potentials have the *same spectrum of eigenvalues*:

$$\text{spec}(V_\ell^{(-)}) = \text{spec}(V_\ell^{(+)}). \quad (206)$$

This is demonstrated via a Darboux operator $\mathcal{A}$ that transforms a solution of the Regge–Wheeler equation into a solution of the Zerilli equation with the same ω :

$$\Psi_\ell^{(+)} = \alpha \mathcal{A} \Psi_\ell^{(-)}, \quad (207)$$

where $\mathcal{A}$ is a first-order differential operator.

Isospectrality has important physical consequences:

- (i) The quasinormal modes (QNMs) of the axial and polar sectors coincide, which is an observationally testable prediction.
- (ii) It suggests the existence of a hidden symmetry of the problem, related to supersymmetry (SUSY) between the two sectors.
- (iii) It is the starting point of *supersymmetric quantum mechanics* (SUSY-QM) applied to black hole perturbations.

H.7 Boundary conditions and quasinormal modes

Equation (199) (and analogously (203)) with boundary conditions:

$$\Psi_\ell^{(-)} \sim e^{-i\omega r_*} \quad (r_* \rightarrow -\infty, \text{ solo entrante al horizonte}), \quad (208)$$

$$\Psi_\ell^{(-)} \sim e^{+i\omega r_*} \quad (r_* \rightarrow +\infty, \text{ solo saliente al infinito}), \quad (209)$$

defines an eigenvalue problem for $\omega \in \mathbb{C}$. The frequencies $\omega_n = \omega_R + i\omega_I$ that simultaneously satisfy both boundary conditions are the *quasinormal modes*.

Properties of QNMs.

- (i) QNM frequencies are *complex*: ω_R is the oscillation frequency and $\omega_I < 0$ is the damping rate (exponential decay in time).
- (ii) The spectrum is *discrete* (although QNMs are not L^2 eigenfunctions in the usual sense, due to the oscillatory behavior at the boundaries).
- (iii) For the Schwarzschild black hole, the dominant QNMs are (in units $M = 1$):

$$\omega_0^{(\ell=2)} \approx 0.3737 - 0.0890i, \quad (210)$$

known to high numerical precision (Leaver's method [6]).

- (iv) For high-overtone modes ($n \gg 1$): $\omega_n \approx \ln 3 / (8\pi M) - i(n + \frac{1}{2}) / 4M$, independent of ℓ .

H.8 Methods for computing the spectrum

H.8.1 WKB approximation

For the barrier potential (201), the third-order WKB approximation gives [7]:

$$\omega^2 \approx V_0 + \left(-2V_0''\right)^{1/2} \Lambda_{\text{WKB}}, \quad (211)$$

where V_0 is the maximum value of the potential, $V_0'' = d^2V/dr_*^2$ evaluated at the maximum, and Λ_{WKB} are WKB corrections computed perturbatively.

For $\ell = 2$ at WKB order:

$$\omega_0^{(\ell=2)} \approx \frac{1}{6\sqrt{3}M} \left(\ell + \frac{1}{2} + \alpha_{\text{WKB}} \right), \quad (212)$$

where α_{WKB} are higher-order corrections.

H.8.2 Leaver's method (continued fractions)

Leaver's exact method [6] expands Ψ in a power series around the horizon:

$$\Psi(r) = e^{i\omega r_*} (r - 2M)^{-i \cdot 4M\omega} \sum_{n=0}^{\infty} a_n \left(\frac{r - 2M}{r} \right)^n, \quad (213)$$

where the coefficients a_n satisfy a three-term recurrence relation. The convergence condition (corresponding to the boundary condition at $r \rightarrow \infty$) is expressed as the requirement that a continued fraction converge, giving an implicit equation for ω that is solved numerically.

H.9 Relation to the Lichnerowicz operator

The relation between the Regge–Wheeler potential and the Lichnerowicz operator is direct. Decomposing the TT perturbation of the black hole into spherical modes:

$$h_{ij}^{\text{TT}}(t, r, \theta, \phi) = \frac{\Psi_\ell^{(-)}(t, r)}{r} \hat{Y}_{ij}^{\ell m}(\theta, \phi), \quad (214)$$

where $\hat{Y}_{ij}^{\ell m}$ are the tensor harmonics of the sphere (eigenfunctions of the operator Δ_L restricted to S^2), the equation $\Delta_L h^{\text{TT}} = 0$ becomes:

$$\left[-\partial_t^2 + \partial_{r_*}^2 - V_\ell^{(-)}(r) \right] \Psi_\ell^{(-)} = 0, \quad (215)$$

which is exactly the Regge–Wheeler equation (199).

This confirms that the **QNM spectrum of the Schwarzschild black hole is the spectrum of the Lichnerowicz operator on that background**, evaluated in the TT sector with quasinormal boundary conditions. The positivity of the potential implies the linear stability of the black hole.

H.10 Linear stability: energetic argument

El agujero negro de Schwarzschild es linealmente estable frente a tensor perturbations: there exist no solutions of the Regge–Wheeler equation with $\text{Im}(\omega) > 0$ (modes growing in time).

Sketch of the energetic argument. Define the conserved energy functional:

$$E[\Psi] = \int_{-\infty}^{+\infty} \left[|\partial_t \Psi|^2 + |\partial_{r_*} \Psi|^2 + V_\ell^{(-)} |\Psi|^2 \right] dr_*, \quad (216)$$

que satisface $\partial_t E = 0$ para soluciones de (199).

Dado que $V_\ell^{(-)} \geq 0$ para todo $r > 2M$ y $\ell \geq 2$ (verifiable directly from (201)), all terms in the integrand of (216) are non-negative.

Si existiera un modo inestable $\Psi \sim e^{\sigma t}$ con $\sigma > 0$, then $\partial_t E = 2\sigma E > 0$, contradicting conservation. Therefore $\sigma \leq 0$ for every bounded solution. $\square$ $\square$

For $\ell = 0$ (spherically symmetric perturbations) and $\ell = 1$ (dipolar perturbations), the analysis is different:

- $\ell = 0$: the most general perturbation corresponds to a mass shift $M \rightarrow M + \delta M$, which is purely a reparametrization and not a dynamical mode.
- $\ell = 1$ (axial): corresponds to adding angular momentum to the black hole ($J = 0 \rightarrow J = \delta J$), also a parameter perturbation, not dynamical.
- $\ell = 1$ (polar): similarly related to a center-of-mass shift.

In all these cases, stability is still guaranteed.

H.11 Fundamental mode and overtones

The QNM spectrum of the axial $\ell = 2$ sector (the observationally dominant one) has the structure:

$$\omega_n^{(\ell)} = \omega_R^{(\ell,n)} - i\omega_I^{(\ell,n)}, \quad n = 0, 1, 2, \dots \quad (217)$$

where $n = 0$ is the *fundamental mode* (longest-lived) and $n > 0$ are the *overtones*. The first numerical values for Schwarzschild ($M = 1$) are:

n	$M\omega_R$	$ M\omega_I $
0	0.37367	0.08896
1	0.34671	0.27392
2	0.30105	0.47828
3	0.25150	0.70514

Table 2: Modos cuasi-normales del sector axial $\ell = 2$ del agujero negro de Schwarzschild (Leaver 1985).

Asymptotic pattern. For $n \rightarrow \infty$ [8]:

$$\omega_n \approx \frac{\ln 3}{8\pi M} - i \frac{2n+1}{4M}, \quad n \gg 1. \quad (218)$$

The real part converges to the universal value $\ln 3/(8\pi M)$, independent of ℓ , a result with possible implications for quantum gravity (area quantization).

H.12 Extension: potentials in Schwarzschild–AdS

For the Schwarzschild–AdS black hole with lapse function:

$$f_{\text{SAdS}}(r) = 1 - \frac{2M}{r} + \frac{r^2}{L^2}, \quad (219)$$

the Regge–Wheeler and Zerilli potentials are modified with additional terms proportional to $1/L^2$:

$$V_{\ell, \text{SAdS}}^{(-)}(r) = f_{\text{SAdS}}(r) \left[\frac{\ell(\ell+1)}{r^2} - \frac{6M}{r^3} + \frac{2}{L^2} \right]. \quad (220)$$

The term $2/L^2$ lifts the potential and contributes to the effective graviton mass in AdS, related to the Breitenlöhner–Freedman bound mentioned in Section 7.

Boundary conditions in AdS. Instead of outgoing radiation at infinity (209), in AdS one imposes Dirichlet conditions (vanishing field at the boundary), discretizing the QNM spectrum into a fully discrete and (for certain mass regimes) real spectrum. This corresponds, via AdS/CFT, to the frequency spectrum of plasma perturbations in the dual field theory.

I Connection to the second-order action: term-by-term derivation

This section complements Section 5.3 by explicitly developing each term of the quadratic action (112) and verifying its form on the Minkowski background (recovering Fierz–Pauli).

I.1 Linearization of $\sqrt{-g} R$ to second order

Using the results of the TFM of Fernández Uria (eqs. 4.12–4.13) generalized to an arbitrary background:

$$\begin{aligned} \sqrt{-g} R \approx \sqrt{-\bar{g}} & \left[\bar{R} + \epsilon \left(\frac{1}{2} h \bar{R} - h^{\mu\nu} \bar{R}_{\mu\nu} + \bar{\nabla}^2 h - \bar{\nabla}^\mu \bar{\nabla}^\nu h_{\mu\nu} \right) \right. \\ & \left. + \epsilon^2 \left(\text{quadratic terms in } h \right) \right]. \end{aligned} \quad (221)$$

I.1.1 The first-order term

The coefficient of ϵ in (221) is:

$$\mathcal{L}^{(1)} = \sqrt{-\bar{g}} \left(\frac{1}{2} h \bar{R} - h^{\mu\nu} \bar{R}_{\mu\nu} + \text{derivada total} \right) = \sqrt{-\bar{g}} h^{\mu\nu} \underbrace{\left(\bar{R}_{\mu\nu} - \frac{1}{2} \bar{g}_{\mu\nu} \bar{R} \right)}_{\bar{G}_{\mu\nu}}. \quad (222)$$

When the background satisfies the Einstein equations, $\bar{G}_{\mu\nu} = 0$, and the first-order term vanishes: $S^{(1)} = 0$.

I.1.2 The second-order term

The coefficient of $\epsilon^2/2$ is the quadratic Lagrangian density. After integrating by parts and discarding boundary terms:

$$\begin{aligned} \mathcal{L}^{(2)} = & -\frac{1}{4} (\bar{\nabla}_\rho h_{\mu\nu}) (\bar{\nabla}^\rho h^{\mu\nu}) + \frac{1}{4} (\bar{\nabla}_\rho h) (\bar{\nabla}^\rho h) \\ & + \frac{1}{2} (\bar{\nabla}_\mu h_{\nu\rho}) (\bar{\nabla}^\nu h^{\mu\rho}) - \frac{1}{2} (\bar{\nabla}_\mu h) (\bar{\nabla}_\nu h^{\mu\nu}) \\ & - \frac{1}{4} \bar{R}_{\mu\rho\nu\sigma} h^{\mu\nu} h^{\rho\sigma} + \frac{1}{2} \bar{R}_{\mu\nu} h^{\mu\rho} h^\nu{}_\rho - \frac{1}{4} \bar{R} (h_{\mu\nu} h^{\mu\nu} - \frac{1}{2} h^2). \end{aligned} \quad (223)$$

I.2 Verification on flat background: recovery of Fierz-Pauli

On the Minkowski background ($\bar{R}_{\mu\nu\rho\sigma} = 0$, $\bar{\nabla} \rightarrow \partial$), all curvature terms vanish and (223) reduces to:

$$\begin{aligned} \mathcal{L}^{(2)} \Big|_\eta = & -\frac{1}{4} (\partial_\rho h_{\mu\nu}) (\partial^\rho h^{\mu\nu}) + \frac{1}{4} (\partial_\rho h) (\partial^\rho h) \\ & + \frac{1}{2} (\partial_\mu h_{\nu\rho}) (\partial^\nu h^{\mu\rho}) - \frac{1}{2} (\partial_\mu h) (\partial_\nu h^{\mu\nu}). \end{aligned} \quad (224)$$

This coincides exactly with the Fierz-Pauli Lagrangian (120) (rescaled by the appropriate factor of κ), confirming that:

$$S_{\text{EH}}^{(2)} \Big|_\eta = S_{\text{FP}}. \quad (225)$$

I.3 Identification with the compact Lichnerowicz form

Integrating (223) by parts, the terms with two derivatives on $h_{\mu\nu}$ reorganize into the Lichnerowicz operator:

$$-\frac{1}{4} (\bar{\nabla}_\rho h_{\mu\nu}) (\bar{\nabla}^\rho h^{\mu\nu}) \rightarrow \frac{1}{4} h^{\mu\nu} (\bar{\square} h)_{\mu\nu} = -\frac{1}{4} h^{\mu\nu} (\Delta_L h)_{\mu\nu} \Big|_{\bar{\square}}, \quad (226)$$

while the cross terms contribute to the Riemann term of Δ_L . The final result is the compact form (117).

Sign verification. The sign of the quadratic action in the TT sector is:

$$S_{\text{TT}}^{(2)} = \frac{1}{4} \int \sqrt{-\bar{g}} h^{\text{TT}\mu\nu} (\Delta_L h^{\text{TT}})_{\mu\nu}, \quad (227)$$

which is *positive definite* when Δ_L has positive spectrum. On the Schwarzschild background, the stability argument of Section H guarantees precisely this for TT modes.

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A Conserved Charges, the Gravitational Energy-Momentum Tensor, and Stability: the Abbott–Deser–Tekin Framework

A.1 The central question: what is $T_{\mu\nu}$ in gravity?

When we linearize the Einstein equations around a background $\bar{g}_{\mu\nu}$, we write (suppressing the cosmological constant for a moment):

$$G_{\mu\nu}^{(1)}[h; \bar{g}] = -\kappa t_{\mu\nu}, \quad (228)$$

where $G_{\mu\nu}^{(1)}$ is the linearized Einstein tensor (48) and $t_{\mu\nu}$ is the perturbation of the matter energy-momentum tensor. A natural question arises: *can we define a gravitational energy-momentum tensor $T_{\mu\nu}^{\text{grav}}$ from the nonlinear terms?*

The answer is subtle. In general relativity, there is no local, covariant, gauge-invariant energy-momentum tensor for the gravitational field. However, one can define *conserved global charges* — energy, momentum, angular momentum — as surface integrals at spatial infinity, provided the background spacetime has appropriate Killing symmetries.

This is the program of Abbott and Deser [12], extended to higher-curvature theories by Deser and Tekin [13]. We now develop this framework in detail.

A.2 The Abbott–Deser decomposition

Consider the Einstein equations with cosmological constant:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = 0. \quad (229)$$

The background $\bar{g}_{\mu\nu}$ satisfies (229) exactly. We write $g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}$ and decompose the left-hand side of (229) as:

$$(G_{\mu\nu} + \Lambda g_{\mu\nu}) = \underbrace{\bar{G}_{\mu\nu} + \Lambda \bar{g}_{\mu\nu}}_{=0} + \underbrace{(G_{\mu\nu} + \Lambda g_{\mu\nu})^{(1)}}_{\text{linear in } h} + \underbrace{(G_{\mu\nu} + \Lambda g_{\mu\nu})^{(\geq 2)}}_{\text{quadratic and higher}}. \quad (230)$$

The gravitational energy-momentum tensor density is defined by grouping *all terms of quadratic and higher order in $h_{\mu\nu}$* on the right-hand side:

$$(G^{\mu\nu} + \Lambda h^{\mu\nu})^{(1)} = -(-\bar{g})^{-1/2} T_{\text{grav}}^{\mu\nu}, \quad (231)$$

where $T_{\text{grav}}^{\mu\nu}$ collects all the nonlinear gravitational self-interaction terms.

Why this definition makes sense. The linear operator on the left of (231) obeys the background Bianchi identity:

$$\mu(G^{\mu\nu} + \Lambda h^{\mu\nu})^{(1)} = 0, \quad (232)$$

which implies the background covariant conservation:

$${}_{\mu}T_{\text{grav}}^{\mu\nu} = 0. \quad (233)$$

This is a covariant conservation law, not an ordinary one. To construct *ordinarily* conserved charges (whose time integrals are constants), we need to contract with a Killing vector.

A.3 Killing vectors and the construction of conserved charges

Let $\bar{\xi}_{\mu}$ be a Killing vector of the background metric, satisfying:

$${}_{\mu}\bar{\xi}_{\nu} + {}_{\nu}\bar{\xi}_{\mu} = 0. \quad (234)$$

Define the current:

$$J^{\mu} \equiv T_{\text{grav}}^{\mu\nu} \bar{\xi}_{\nu}. \quad (235)$$

Using (233) and the Killing equation (234):

$${}_{\mu}J^{\mu} = ({}_{\mu}T_{\text{grav}}^{\mu\nu})\bar{\xi}_{\nu} + T_{\text{grav}}^{\mu\nu}({}_{\mu}\bar{\xi}_{\nu}) = 0 + \frac{1}{2}T_{\text{grav}}^{\mu\nu}({}_{\mu}\bar{\xi}_{\nu} + {}_{\nu}\bar{\xi}_{\mu}) = 0. \quad (236)$$

Since $T_{\text{grav}}^{\mu\nu}$ is symmetric, only the symmetric part of ${}_{\mu}\bar{\xi}_{\nu}$ contributes, which vanishes by the Killing equation. Now, ${}_{\mu}J^{\mu} = \partial_{\mu}(\sqrt{-\bar{g}} J^{\mu})/\sqrt{-\bar{g}}$, so the quantity $\sqrt{-\bar{g}} J^{\mu}$ has an *ordinary* divergence that vanishes:

$$\partial_{\mu}(\sqrt{-\bar{g}} T_{\text{grav}}^{\mu\nu} \bar{\xi}_{\nu}) = 0. \quad (237)$$

This gives the conserved charge:

$$E(\bar{\xi}) = \frac{1}{8\pi G} \int_{\Sigma} d^3x \sqrt{-\bar{g}} T_{\text{grav}}^{0\nu} \bar{\xi}_{\nu}, \quad (238)$$

which is the *Killing energy* (or Killing charge) associated to $\bar{\xi}^{\mu}$. When $\bar{\xi}^{\mu}$ is a timelike Killing vector, this is the conserved energy of the gravitational field.

A.4 The superpotential and flux-integral form

The key result of Abbott and Deser is that (238) can be written as a *two-dimensional surface integral* — a “flux integral” at spatial infinity. This is the gravitational analogue of Gauss’s law.

A.4.1 The trace-reversed perturbation

Define the *trace-reversed perturbation*:

$$H^{\mu\nu} \equiv h^{\mu\nu} - \frac{1}{2}\bar{g}^{\mu\nu}h, \quad h \equiv \bar{g}^{\mu\nu}h_{\mu\nu}. \quad (239)$$

Note that $H^{\mu}_{\mu} = -h$ (for $D = 4$: $H = -h$, sign flip), and that $H^{\mu\nu}$ is the gravitational analogue of the trace-reversed metric perturbation used in linearized general relativity.

A.4.2 The Abbott–Deser superpotential

The gravitational energy-momentum tensor can be written as:

$$(-\bar{g})^{-1/2}T^{\mu\nu} = {}_{\alpha\beta}K^{\mu\alpha\nu\beta} + X^{\mu\nu}, \quad (240)$$

where the *superpotential* $K^{\mu\alpha\nu\beta}$ is defined by:

$$K^{\mu\alpha\nu\beta} = \frac{1}{2}[\bar{g}^{\mu\beta}H^{\nu\alpha} + \bar{g}^{\nu\alpha}H^{\mu\beta} - \bar{g}^{\mu\nu}H^{\alpha\beta} - \bar{g}^{\alpha\beta}H^{\mu\nu}], \quad (241)$$

and has the algebraic symmetries of the Riemann tensor:

$$K^{\mu\alpha\nu\beta} = K^{\nu\beta\mu\alpha} = -K^{\alpha\mu\nu\beta} = -K^{\mu\alpha\beta\nu}. \quad (242)$$

The remaining term $X^{\mu\nu}$ is:

$$X^{\mu\nu} = \frac{1}{2}[\bar{R}^{\mu}{}_{\alpha\beta}{}^{\nu}H^{\alpha\beta} - \Lambda H^{\mu\nu}], \quad (243)$$

which can be written equivalently as $X^{\mu\nu} = \frac{1}{2}\bar{R}^{\nu}{}_{\lambda\alpha\beta}K^{\mu\lambda\alpha\beta}$ using the background equations.

A.4.3 Converting to a flux integral

Multiplying (240) by the Killing vector $\bar{\xi}_{\nu}$ and using the Killing vector identity:

$${}_{\beta\alpha}\bar{\xi}_{\nu} = \bar{R}^{\lambda}{}_{\beta\alpha\nu}\bar{\xi}_{\lambda}, \quad (244)$$

one shows that $(-\bar{g})^{-1/2}T^{\mu\nu}\bar{\xi}_{\nu}$ is a total divergence of an antisymmetric tensor density $\mathcal{F}^{\mu\nu}$:

$$(-\bar{g})^{-1/2}T^{\mu\nu}\bar{\xi}_{\nu} = {}_{\alpha}\mathcal{F}^{\mu\alpha}, \quad \mathcal{F}^{\mu\alpha} = -\mathcal{F}^{\alpha\mu}. \quad (245)$$

Since the divergence of an antisymmetric tensor density is an ordinary derivative, we get:

$$\partial_{\alpha}(\sqrt{-\bar{g}}\mathcal{F}^{\mu\alpha}) = 0, \quad (246)$$

and the Killing energy becomes a surface integral:

$$E(\bar{\xi}) = \frac{1}{8\pi G} \oint dS_i \sqrt{-\bar{g}} [{}_{\beta}K^{0i\nu\beta} - K_j^{0j\nu i}] \bar{\xi}_{\nu}, \quad (247)$$

where the integral is over a closed two-dimensional spatial surface at infinity.

Flat-space limit. When $\Lambda = 0$ and $\bar{g}_{\mu\nu} = \eta_{\mu\nu}$, the Killing vector for time translations is $\bar{\xi}^{\mu} = (1, \mathbf{0})$, and (247) reduces to the ADM mass formula [14]:

$$E_{\text{ADM}} = \frac{1}{16\pi G} \oint dS_i [\partial_j h_{ij} - \partial_i h_{jj}]. \quad (248)$$

A.5 The role of $T_{\mu\nu}$: what it is and what it is not

At this point we can give a precise answer to the question raised at the beginning of this section.

What $T_{\mu\nu}^{\text{grav}}$ is. It is the *effective source* that the nonlinear gravitational self-interactions create in the linearized field equations. In the perturbation expansion:

$$G_{\mu\nu}^{(1)}[h; \bar{g}] = -\kappa(T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{\text{grav}}), \quad (249)$$

the matter perturbation $t_{\mu\nu}$ and the gravitational self-energy $T_{\mu\nu}^{\text{grav}}$ both source the linearized metric perturbation $h_{\mu\nu}$. The gravitational self-energy is quadratic (and higher) in $h_{\mu\nu}$:

$$(-\bar{g})^{1/2} T_{\text{grav}}^{\mu\nu} \sim h \cdot h, \quad h \cdot \partial h, \quad \partial h \cdot \partial h, \dots \quad (250)$$

What $T_{\mu\nu}^{\text{grav}}$ is not.

- (i) It is *not* locally gauge-invariant. Under $h_{\mu\nu} \rightarrow h_{\mu\nu} + {}_{\mu}\xi_{\nu} + {}_{\nu}\xi_{\mu}$, the tensor $T_{\mu\nu}^{\text{grav}}$ changes. Only its integral over a closed surface (the Killing charge) is gauge-invariant.
- (ii) It is *not* uniquely defined. One can add terms that are identically conserved (${}^{\mu\nu}F_{\mu\nu}$ for any antisymmetric $F_{\mu\nu}$) without changing the conserved charge. The superpotential $K^{\mu\alpha\nu\beta}$ is not unique either.
- (iii) It is *not* a tensor under arbitrary coordinate transformations of the full spacetime — it transforms as a tensor only under the isometries of the background.

Physical interpretation. The conserved Killing charge $E(\bar{\xi})$ is the physically meaningful quantity. For a timelike Killing vector, it plays the role of energy. Its positivity — which is the content of the positive energy theorem — is what guarantees stability.

A.6 Stability from positivity of the Killing energy

The background $\bar{g}_{\mu\nu}$ is linearly stable if and only if the Killing energy $E(\bar{\xi})$ is *non-negative* for all perturbations $h_{\mu\nu}$ that satisfy the linearized equations of motion and the appropriate boundary conditions.

Connection to the quadratic action. There is a direct link between the Killing energy and the quadratic action $S^{(2)}[h]$. In the canonical (ADM) decomposition, the Killing energy for the background timelike Killing vector is:

$$E(\bar{\xi}) = - \int d^3x \bar{N} \bar{g}^{-1/2} \mathcal{H}, \quad (251)$$

where $\mathcal{H}$ is the Hamiltonian density evaluated at second order in the perturbation. This is precisely what Abbott and Deser compute in their Section 4 for de Sitter and anti-de Sitter backgrounds. The positivity of $E(\bar{\xi})$ is equivalent to the positive definiteness of the kinetic operator in $S^{(2)}$.

A.7 De Sitter space: stability restricted by the horizon

A.7.1 The de Sitter metric and its Killing vector

The de Sitter metric in flat slicing coordinates is:

$$ds^2 = -dt^2 + f^2(t) d\mathbf{x}^2, \quad f(t) = e^{\sqrt{\Lambda/3}t}, \quad \Lambda > 0. \quad (252)$$

The relevant Killing vector for energy is:

$$\bar{\xi}^\mu = (-1, \sqrt{\frac{\Lambda}{3}} \mathbf{x}), \quad (253)$$

with norm:

$$\bar{\xi}^2 = -1 + \frac{\Lambda}{3} f^2 |\mathbf{x}|^2. \quad (254)$$

This is timelike only in the region $\frac{\Lambda}{3} f^2 |\mathbf{x}|^2 < 1$, i.e., *inside the de Sitter event horizon*:

$$|\mathbf{x}| < \frac{1}{f} \sqrt{\frac{3}{\Lambda}}. \quad (255)$$

Outside the horizon, $\bar{\xi}^\mu$ becomes spacelike and the Killing energy loses its interpretation as an energy.

A.7.2 Small fluctuations in de Sitter: TT sector

In the TT gauge (transverse-traceless), Abbott and Deser introduce canonical variables:

$$Q_{ij} = \sqrt{\frac{1}{2}} f^{-2} h_{ij}, \quad P^{ij} = \sqrt{2} f^2 (p^{ij} - 2\sqrt{\frac{\Lambda}{3}} f^{-1} h^{ij}), \quad (256)$$

in terms of which the Hamiltonian density takes the manifestly positive form:

$$\mathcal{H} = -T^0_0 = \frac{1}{2} [f^{-3} (P^{ij})^2 + f (\nabla Q_{ij})^2]. \quad (257)$$

The Killing energy density is:

$$T^{0\nu} \bar{\xi}_\nu = \frac{1}{2} [f^{-3} (P^{ij})^2 + f (\nabla Q_{ij})^2] - \sqrt{\frac{\Lambda}{3}} x_i P^{jk} \partial_i Q_{jk}. \quad (258)$$

Using the triangle inequality (as in the scalar field analogy of Abbott–Deser Section 3), one shows:

$$T^{0\nu} \bar{\xi}_\nu \geq 0 \quad \Longleftrightarrow \quad \frac{\Lambda}{3} f^2 |\mathbf{x}|^2 < 1, \quad (259)$$

which is precisely the condition (255). Therefore:

De Sitter space is linearly stable against TT gravitational perturbations *inside the event horizon*. Outside the horizon, the Killing energy can be negative, which is the gravitational manifestation of Hawking radiation from the cosmological horizon.

A.8 Anti-de Sitter space: global stability

A.8.1 The AdS metric and global timelike Killing vector

Anti-de Sitter space corresponds to $\Lambda < 0$. A complete metric is:

$$ds^2 = -\cosh^2\left(\sqrt{\frac{|\Lambda|}{3}} r\right) dt^2 + dr^2 + \sinh^2\left(\sqrt{\frac{|\Lambda|}{3}} r\right) d\Omega_2^2. \quad (260)$$

The Killing vector $\bar{\xi}^\mu = (-1, \mathbf{0})$ is *globally timelike*:

$$\bar{\xi}^2 = -\cosh^2\left(\sqrt{\frac{|\Lambda|}{3}} r\right) < 0 \quad \forall r \geq 0. \quad (261)$$

There is no event horizon in AdS, so the Killing energy is well-defined everywhere.

A.8.2 Hamiltonian in anti-de Sitter

For the anti-de Sitter background, ${}^{(3)}\bar{R} = 2\Lambda < 0$ and the background momentum vanishes $\bar{\pi}^{ij} = 0$. The Hamiltonian density to second order in the TT sector is [12]:

$$\mathcal{H}_{\text{AdS}} = -T^0_0 = \bar{N}\bar{g}^{-1/2} \left[p_{ij}p^{ij} + \bar{g} \left(\frac{1}{4} (\bar{D}_k h_{ij})^2 + \frac{|\Lambda|}{6} (h_{ij})^2 \right) \right], \quad (262)$$

which is manifestly positive definite ($\bar{N} > 0$, $\bar{g} > 0$, $|\Lambda| > 0$). Every term is non-negative. Since $\bar{\xi}^i = 0$, the Killing energy is simply:

$$E(\bar{\xi}) = -\frac{1}{8\pi G} \int d^3x T^0_0 \geq 0. \quad (263)$$

This proves:

Anti-de Sitter space is linearly stable against all TT gravitational perturbations. The Killing energy is strictly non-negative, and vanishes only for the background itself. Moreover, the Hamiltonian contains a “mass-like” term $\propto |\Lambda|(h_{ij})^2$, but this does not make the graviton massive — the TT sector still has only two physical degrees of freedom.

Why the $|\Lambda|(h_{ij})^2$ term does not give mass. In flat space, a mass term $m^2 h_{\mu\nu} h^{\mu\nu}$ breaks the gauge invariance and gives the graviton five degrees of freedom (Fierz–Pauli with $m \neq 0$). In AdS, the term $|\Lambda|(h_{ij})^2$ appears in the Hamiltonian but is *not* a gauge-breaking mass term — it arises from the background curvature coupling $\bar{R}_{i\mu j\nu} h^{\mu\nu}$ in the Lichnerowicz operator. The TT gauge is still consistent, and the graviton remains massless with two physical degrees of freedom.

A.9 Linearization in AdS: explicit computation

We now work out explicitly the linearization of the gravitational equations around the AdS_D background, connecting the results of Sections 2–3 to the Abbott–Deser formalism.

A.9.1 The AdS_D background

The D -dimensional anti-de Sitter space with cosmological constant $\Lambda = -(D-1)(D-2)/(2L^2)$ has:

$$\bar{R}_{\mu\nu\rho\sigma} = \frac{\Lambda}{(D-1)(D-2)/2} (\bar{g}_{\mu\rho}\bar{g}_{\nu\sigma} - \bar{g}_{\mu\sigma}\bar{g}_{\nu\rho}) = -\frac{1}{L^2} (\bar{g}_{\mu\rho}\bar{g}_{\nu\sigma} - \bar{g}_{\mu\sigma}\bar{g}_{\nu\rho}), \quad (264)$$

$$\bar{R}_{\mu\nu} = \frac{2\Lambda}{D-2} \bar{g}_{\mu\nu} = -\frac{D-1}{L^2} \bar{g}_{\mu\nu}, \quad (265)$$

$$\bar{R} = \frac{2D\Lambda}{D-2} = -\frac{D(D-1)}{L^2}. \quad (266)$$

The background satisfies the Einstein equations with Λ : $\bar{G}_{\mu\nu} + \Lambda\bar{g}_{\mu\nu} = 0$.

A.9.2 Lichnerowicz operator in AdS_D

Substituting (264) and (265) into the definition of the Lichnerowicz operator (45):

$$\begin{aligned} (\Delta_L h)_{\mu\nu} &= -\bar{\square} h_{\mu\nu} + \frac{2}{L^2} (\bar{g}_{\mu\rho}\bar{g}_{\nu\sigma} - \bar{g}_{\mu\sigma}\bar{g}_{\nu\rho}) h^{\rho\sigma} - \frac{D-1}{L^2} (h_{\mu\nu} h_{\rho\rho} + h_{\nu\rho} h_{\mu\rho}) \\ &= -\bar{\square} h_{\mu\nu} + \frac{2}{L^2} (h_{\mu\nu} - \bar{g}_{\mu\nu} h) + \frac{2(D-1)}{L^2} h_{\mu\nu} \\ &= \left(-\bar{\square} + \frac{2D}{L^2} \right) h_{\mu\nu} - \frac{2}{L^2} \bar{g}_{\mu\nu} h. \end{aligned} \quad (267)$$

On the TT sector ($h = 0, {}^\nu h_{\mu\nu} = 0$), this simplifies to:

$$\boxed{(\Delta_L h^{\text{TT}})_{\mu\nu} \Big|_{\text{AdS}_D} = \left(-\bar{\square} + \frac{2D}{L^2} \right) h_{\mu\nu}^{\text{TT}}.} \quad (268)$$

The effective “mass” of the graviton in AdS is $m_{\text{eff}}^2 = 2D/L^2$, which is positive and above the Breitenlöhner–Freedman bound $m_{\text{BF}}^2 = -(D-1)^2/(4L^2)$ for spin-2 fields.

A.9.3 Linearized Einstein equations in AdS_D

The linearized Einstein equations $G_{\mu\nu}^{(1)} + \Lambda h_{\mu\nu} = 0$ in the TT gauge become:

$$\begin{aligned} -\frac{1}{2}(\Delta_L h^{\text{TT}})_{\mu\nu} + \Lambda h_{\mu\nu}^{\text{TT}} &= 0 \\ \frac{1}{2} \left(\bar{\square} - \frac{2D}{L^2} \right) h_{\mu\nu}^{\text{TT}} - \frac{(D-1)(D-2)}{2L^2} h_{\mu\nu}^{\text{TT}} &= 0 \\ \left(\bar{\square} - \frac{2(D-1)}{L^2} \right) h_{\mu\nu}^{\text{TT}} &= 0. \end{aligned} \quad (269)$$

This can be written as:

$$\boxed{\left(\bar{\square}_{\text{AdS}} - M_{\text{grav}}^2 \right) h_{\mu\nu}^{\text{TT}} = 0, \quad M_{\text{grav}}^2 = \frac{2(D-1)}{L^2} = \frac{2|\Lambda|D}{(D-2)}.} \quad (270)$$

This is a Klein–Gordon-type equation for a spin-2 field with an effective mass set by the cosmological constant. This is the AdS generalization of the flat-space wave equation $\square h^{\text{TT}} = 0$.

A.9.4 Mode decomposition and spectrum in AdS_D

In AdS_D with radius L , we expand the TT graviton in tensor spherical harmonics. For the static patch of AdS_4 (relevant for boundary conditions), the modes satisfy:

$$\left[\partial_t^2 - \partial_{r_*}^2 + V_{\text{AdS}}(r) \right] \Psi = 0, \quad (271)$$

where r_* is the tortoise coordinate and the effective potential is:

$$V_{\text{AdS}}(r) = f(r) \left[\frac{\ell(\ell+2)}{r^2} + \frac{2}{L^2} \right], \quad f(r) = 1 + \frac{r^2}{L^2}, \quad (272)$$

for tensor modes with angular momentum $\ell \geq 2$. The additional term $2/L^2$ (compared to flat space) lifts the potential and ensures that all modes have positive “energy squared” ω^2 , confirming stability.

Boundary conditions in AdS. Unlike asymptotically flat spacetimes where outgoing radiation at infinity is imposed, in AdS the boundary at $r \rightarrow \infty$ is timelike. Two choices of boundary conditions are physically relevant:

- **Dirichlet:** $h_{\mu\nu}^{\text{TT}} \rightarrow 0$ as $r \rightarrow \infty$. Leads to a discrete spectrum of normal modes.
- **Neumann** (or mixed): allows non-normalizable modes relevant for AdS/CFT, where the boundary value of $h_{\mu\nu}$ sources the stress-energy tensor of the dual CFT.

A.10 Extension to higher-curvature theories: Deser–Tekin charges

A.10.1 Generic higher-curvature gravity

Consider a generic gravity theory with action:

$$I = \int d^D x \sqrt{-g} \mathcal{L}(g_{\mu\nu}, R_{\mu\nu\rho\sigma}), \quad (273)$$

which admits a constant-curvature vacuum $\bar{g}_{\mu\nu}$ (possibly without an explicit Λ in the action, as the vacuum curvature is generated dynamically). The field equations are:

$$\mathcal{F}_{\mu\nu}(g, R, \nabla R, R^2, \dots) = \kappa t_{\mu\nu}, \quad (274)$$

where $\mathcal{F}_{\mu\nu}$ is the “Einstein tensor” of the theory (obtained by varying the action with respect to $g^{\mu\nu}$).

A.10.2 The Deser–Tekin procedure

Following Deser and Tekin [13], we decompose $g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}$ and write:

$$\mathcal{F}_{\mu\nu} = \underbrace{\mathcal{F}_{\mu\nu}(\bar{g})}_{=0} + \underbrace{\mathcal{O}(\bar{g})_{\mu\nu}{}^{\rho\sigma} h_{\rho\sigma}}_{\text{linear operator}} + \kappa T_{\mu\nu}, \quad (275)$$

where $T_{\mu\nu}$ collects all nonlinear terms (including matter). The crucial observation is that the linear operator $\mathcal{O}(\bar{g})$ inherits the background Bianchi identity:

$${}^\mu \mathcal{O}(\bar{g})_{\mu\nu}{}^{\rho\sigma} = 0, \quad (276)$$

regardless of the specific gravity theory. This guarantees that ${}^\mu T_{\mu\nu} = 0$, and the same Killing-vector construction goes through to define conserved charges.

A.10.3 Quadratic gravity: the general charge formula

For the quadratic gravity action (273) with $\mathcal{L} = R/\kappa + \alpha R^2 + \beta R_{\mu\nu}^2 + \gamma \mathcal{G}$ (where $\mathcal{G} = R_{\mu\nu\rho\sigma}^2 - 4R_{\mu\nu}^2 + R^2$ is the Gauss–Bonnet term), the conserved Killing charges are:

$$Q^\mu(\bar{\xi}) = \mathcal{C} \int d^{D-1}x \sqrt{-\bar{g}} \bar{\xi}_\nu G^{(1)\mu\nu} + (2\alpha + \beta) \int dS_i \sqrt{-\bar{g}} \mathcal{J}^{\mu i} [\partial R] + \beta \int dS_i \sqrt{-\bar{g}} \mathcal{K}^{\mu i} [\partial G^{(1)}], \quad (277)$$

where the effective coupling is:

$$\mathcal{C} = -\frac{1}{\kappa} + \frac{8\Lambda D}{(D-2)^2} \left(D\alpha + \frac{\beta}{D-1} \right) + \frac{4\Lambda\gamma(D-3)(D-4)}{(D-1)(D-2)}, \quad (278)$$

and $\mathcal{J}, \mathcal{K}$ are surface terms arising from the $\nabla^2 R$ and $\nabla^2 G^{(1)}$ contributions.

Key feature. The first term in (277) is the Abbott–Deser piece (with a modified coupling), and the additional surface integrals come from the higher-derivative terms in the action. For solutions that are asymptotically (A)dS at least as fast as Schwarzschild–AdS (i.e., $h_{\mu\nu} \sim r^{-(D-3)}$ at large r), the $\mathcal{J}$ and $\mathcal{K}$ terms vanish asymptotically, and the charge reduces to:

$$E = \mathcal{C} \cdot \frac{D-2}{4G_D} r_0^{D-3}, \quad (279)$$

where r_0 is the Schwarzschild parameter.

Sign of the energy and stability. The sign of $\mathcal{C}$ determines whether the energy is positive or negative. For Einstein–Gauss–Bonnet theory ($\alpha = \beta = 0, \gamma \neq 0$), the effective coupling flips sign relative to pure Einstein theory, but the asymptotically AdS black hole solution also has a flipped sign in $h_{\mu\nu}$, and the two sign flips compensate, giving $E > 0$. This is the main result of [13]:

In Einstein–Gauss–Bonnet theory, both the flat branch and the AdS branch of asymptotically-(A)dS solutions have positive energy. The AdS vacuum is a stable zero-energy state.

A.10.4 The zero-energy model

An interesting special case is the theory with action $I = \int d^D x \sqrt{-g} \tilde{R}^{\mu\nu} \tilde{R}_{\mu\nu}$, where $\tilde{R}_{\mu\nu} = R_{\mu\nu} - (R/D)g_{\mu\nu}$ is the traceless Ricci tensor. For this theory, $D\alpha + \beta = 0$ (in the parametrization (278)), which gives $\mathcal{C} = 0$ and hence $E = 0$ for *all* asymptotically (A)dS solutions of all sizes. Creating arbitrarily large black holes costs no energy in this theory — the moduli space of solutions is continuously degenerate.

A.11 Summary: the two-step procedure for stability analysis

We can now summarize the complete program for analyzing stability of a gravitational background $\bar{g}_{\mu\nu}$:

- Step 1. Identify the background and its symmetries.** Find a solution $\bar{g}_{\mu\nu}$ to the full (nonlinear) field equations and identify its Killing vectors $\bar{\xi}_{(a)}^\mu$.
- Step 2. Linearize the field equations.** Compute $G_{\mu\nu}^{(1)}[h; \bar{g}]$ (or $\mathcal{F}_{\mu\nu}^{(1)}$ for higher-curvature theories) using the formalism of Sections 2–3.
- Step 3. Define $T_{\mu\nu}^{\text{grav}}$.** Group all nonlinear terms as the gravitational energy-momentum tensor via (231).
- Step 4. Construct conserved Killing charges.** Use the superpotential (241) and the flux integral (247) to write the Killing energy as a surface integral.
- Step 5. Check positivity.** For small fluctuations, use the Hamiltonian form (251) to check whether $E(\bar{\xi}) \geq 0$. For all fluctuations (including large ones), use supergravity techniques (Killing spinors) as in the $\Lambda < 0$ case.
- Step 6. Interpret the result.** If $E(\bar{\xi}) \geq 0$ for all admissible perturbations, the background is stable. If there exist perturbations with $E < 0$, the system is unstable — it can lower its energy by exciting those modes.

Steps 2–5 are intimately connected to the spectrum of the Lichnerowicz operator Δ_L . In the TT sector:

- $E(\bar{\xi}) \geq 0 \iff \text{spec}(\Delta_L)|_{\text{TT}} \geq 0$.
- A negative eigenvalue of Δ_L in the TT sector signals a tachyonic instability (Gregory–Laflamme type for black strings, or bulk instability for Kaluza–Klein compactifications).
- The Breitenlöhner–Freedman bound m_{BF}^2 is precisely the *minimum* eigenvalue of Δ_L in AdS that is still consistent with stability (fields with $m^2 \geq m_{\text{BF}}^2$ are stable despite having $m^2 < 0$ in the usual sense).